



SAPIENZA
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Exploitation of Sentinel2 Multi-Temporal Dataset to monitor effusive volcanic Activity : an Application to Stromboli Volcano

Facoltà di Ingegneria Civil e Industriale
Dipartimento di Ingegneria Meccanica e Aerospaziale
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Nivetha Rajkumar
Matricola 1940472

Relatore
Maria Antonietta Marsella

Correlatore
Anna Maria Gargiulo

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Abstract

Lava flow mapping has direct relevance to volcanic hazards monitoring and risk assessment. Satellite remote sensing of volcanoes is very useful because it provides a comprehensive dataset covering large areas through a variety of acquisition modes ranging from visible to infra-red and radar. Satellite remote sensing techniques are increasingly used to map newly erupted lava, thanks to their capability to survey large areas with frequent revisit time and accurate spatial resolution. Satellite sensing can also access remote locations and hazardous regions without difficulty. Visible and infrared satellite data are routinely used to detect the distributions of volcanic deposits and monitor thermal features. However, clouds are a serious obstacle for optical sensors since they cannot be penetrated by optical radiation. On the other hand, radar satellite data have been playing an important role in surface change detection and image classification, being able to operate in all weather conditions, although their use is hampered by the special imaging geometry, the complicated scattering process, and the presence of speckle noise. Thus, optical and radar data are complementary data sources that can be used to map lava flows effectively, in addition to mitigating cloud obstruction and improving change detection performance. This report focuses on the comprehensive mapping and analysis of lava flows eruptions of Stromboli, a volcano situated in the Mediterranean region, well-known for its persistent eruptive activity. In this work eruptions occurred between 2019 and 2023 are considered by analyzing the Sentinel-2 Multispectral Instrument (MSI) satellite data. The study aims to provide insights into the spatial and temporal dynamics of lava flows, their extent, and their impact on the surrounding environment. We use Sentinel-2 MSI satellite imagery to investigate the Stromboli volcanic activities in terms of eruption. A spatial resolution of 20 m/pixel in the SWIR bands, makes Sentinel-2 data an excellent tool for detecting small, high-temperature thermal anomalies from active hot targets that enhances potential for volcanic monitoring. The process includes sentinel-2 image pre-processing such as converting all the bands of MSI image to the same resolution (10m/pixels) and acquiring datasets containing strictly the required bands to calculate thermal anomalies. The lava flow mapping is then performed by evaluating and deriving an optimal criterion which can be inferred from the spectrum of the pixels corresponding to the lava flows. Finally, the outputs obtained on hotspot and lava flow detection have been analyzed and validated through the adaptation of an existing Machine Learning methodology based on Sentinel-2 multispectral imagery and developed on Google Earth Engine.

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1 Introduction

Satellite thermal remote sensing deals with the acquisition, processing and interpretation of satellite data for study and monitoring of volcanic behaviors and related hazards. Using satellite observations, pre-eruptive parameters in the form of ground deformation (change in shape) and changes in surface temperature can be measured. This detection of deformation and thermal anomalies will improve understanding of the significance of detected deformation and thermal anomalies for disaster risk management. There are a number of Earth Observation Satellites that provide these data on a free, full and open basis. The Satellite datum that are vastly used for various earth observation purposes are studied and displayed as below.

The Initial step was to be involved in an extensive study to understand the existing journals, papers and other references relevant to earth observation and remote sensing activities to refine the research question. Upon developing a clear and concise idea broad enough to allow for exploration, it was possible to outline the objectives, methodologies and expected outcomes of the projected Stromboli volcano monitoring. Now, having all the required knowledge, the next step was to choose the right Volcano imagery, comparing the available sources and opting for the Sentinel-2 data based on various criteria. Then the Sentinel-2 data has been downloaded for image processing for an available period of 2017 to 2023 September from the Copernicus open access hub. These images were then subjected to pre-processing using ESA's SNAP tool where the images have been studied to understand the SWIR bands spectral intensities and Thermal anomalies specific to Stromboli. By varying the different SWIR band values to define a band math that can map accurately the high intensity and Mid-low intensity pixels which can then be imported as a raster data on QGIS in order to get the lava flow areas and other geographical data to fulfill the research objective. Finally, the obtained results have been compared and verified with a Google Earth Engine's API.

1.1 Objective of the Study

The primary objective of this study is to conduct a comprehensive and detailed mapping of lava flows from Stromboli volcano eruptions using satellite imagery, specifically focusing on Sentinel-2 MSI data. This research aims to achieve the following goals:

Accurate Mapping of Lava Flow Extent: Utilize advanced remote sensing techniques to accurately map the extent of lava flows resulting from Stromboli volcano eruptions. This involves identifying, delineating, and analyzing the spatial distribution of lava flows over the specified period (2017-2023).

Temporal Analysis of Lava Flow Dynamics: Investigate the temporal evolution of lava flows to understand the patterns and variations in their behavior over time. Analyze multiple Sentinel-2 MSI images spanning the study period to detect changes in lava flow morphology and volume.

Identification of Eruption Patterns: Identify recurring eruption patterns and trends by correlating lava flow data with historical eruption records. This includes categorizing different types of eruptions based on their intensity, duration, and impact on the surrounding environment.

Assessment of Environmental Impact: Evaluate the environmental impact of lava flows on the local ecosystem, infrastructure, and communities residing in proximity to Stromboli volcano. Assess changes in land cover, vegetation, and topography caused by lava flows and analyze their implications for the environment.

By achieving these objectives, this study intends to provide valuable insights into the behavior of Stromboli volcano's lava flows, enhance our understanding of volcanic processes, and contribute to the development of strategies for mitigating the risks associated with volcanic eruptions.

1.2 Stromboli Volcano: Geology and Eruption Characteristics

Stromboli, situated in the Tyrrhenian Sea, stands as one of Italy's most active and distinctive volcanoes. A comprehensive understanding of its geology and eruption characteristics is paramount for assessing the potential risks it poses and for effective volcanic risk management.

Stromboli is a stratovolcano, characterized by its conical shape formed through the accumulation of various volcanic materials, including lava flows, ash, and volcanic rocks. It is part of the Aeolian Islands, a volcanic arc shaped by the subduction of the African Plate beneath the Eurasian Plate.

The island's geological composition encompasses diverse lava types, primarily basaltic andesite, and andesite, indicating a complex magmatic system beneath the volcano. The presence of ancient and recent lava flows has significantly influenced the island's topography, offering valuable insights into the volcano's historical activity.

Stromboli is renowned for its nearly continuous mild explosive activity. What sets Stromboli apart is its characteristic Strombolian explosions, which involve the frequent ejection of incandescent lava fragments. These explosions result from the release of gas bubbles from the magma, creating captivating lava fountains.

The volcano's activity is categorized into distinct levels, ranging from mild explosions to more intense and potentially hazardous eruptions. These Strombolian explosions often accompany the emission of volcanic ash and pyroclastic material, posing risks to the local populace and infrastructure.

In addition to Strombolian explosions, Stromboli experiences occasional effusive eruptions, where lava flows descend the volcano's slopes. These lava flows pose direct threats to nearby settlements, making their monitoring crucial for hazard assessment and evacuation planning.

2 State of the Art

The "state of the art" refers to the most recent and advanced level of technology, knowledge, or development in a particular field. In our case, this involves not just a summary of previous studies but an analytical and critical evaluation that forms the foundation to the thesis.

2.1 Remote Sensing in Volcanology

Remote sensing technology has revolutionized the field of volcanology, offering valuable tools for monitoring, analyzing, and understanding volcanic activities. Through the use of various remote sensing techniques, scientists can gather critical data without direct physical contact with the volcanic site, enhancing safety and providing a broader perspective on volcanic processes.

Satellite-based remote sensing has become a cornerstone in volcanic research. Satellites equipped with multispectral sensors, such as Sentinel-2, can capture data across different wavelengths. This capability allows scientists to monitor volcanic activities including changes in land temperature, gas emissions, and lava flow dynamics. Continuous and wide-area coverage of satellite imagery aids in the early detection of potential eruptions and provides crucial information for hazard assessment and emergency response planning.

Thermal imaging plays a significant role in detecting subtle temperature variations on a volcano's surface. Thermal cameras and sensors can identify thermal anomalies, indicating areas of increased heat associated with magma movement or vent activity. These anomalies are pivotal in pinpointing potential eruption sites and assessing the level of volcanic activity. Thermal imaging provides a real-time view of the heat distribution, aiding volcanologists in making informed decisions.

Light Detection and Ranging (LIDAR) technology utilizes laser pulses to create highly detailed and precise three-dimensional maps of the Earth's surface. In the context of volcanology, LIDAR is invaluable for mapping terrain changes resulting from eruptions. It can detect even subtle alterations in the landscape, such as ground deformation or the formation of new vents. The high-resolution data produced by LIDAR is essential for accurate hazard assessments, eruption modeling, and understanding the long-term geological impacts of volcanic activities.

In conclusion, remote sensing techniques provide a non-intrusive, comprehensive, and timely means of monitoring volcanic activities. These methods not only enhance our understanding of volcanic processes but also contribute significantly to mitigating risks and ensuring the safety of communities living in volcanic regions.

2.2 Earth Observation Satellites for Volcano monitoring

The most adopted Satellites for Volcano monitoring are briefly described

Copernicus Sentinel-1 twin satellites operate day and night performing C-band synthetic aperture radar SAR imaging, acquiring imagery irrespective of the weather. Differential SAR interferometry (DInSAR) techniques is one of the applications volcanologists most widely used to study the deformation phenomena with centimeter to millimeter accuracy.

Copernicus Sentinel-2 satellites provide Thermal remote sensing data to study and monitor volcanic activity. The availability of 13 bands, in particular a spatial resolution of 20 m/pixel in the SWIR (Short-wave Infrared) bands help monitor the thermal emission of high-temperature bodies. The Sentinel-2 has three spatial resolutions of 10m, 20m, 60m.

The thermal infrared (TIR) channels of Sentinel-3's SLSTR (Sea and Land Surface Temperature Radiometer) is used to monitor volcanic ash during both day and night.

Whereas Sentinel-5P's TROPOMI instrument has UV Channels that can be exploited in order to study the measure of several trace gases in the lower atmosphere: for instance, nitrogen dioxide (NO_2), ozone(O_3), formaldehyde (CH_2O), sulphur dioxide (SO_2), methane (CH_4), carbon monoxide (CO) and aerosols.

SATELLITE	SENSOR/WAVELENGTH		MONITORING PARAMETER
Sentinel-1	SAR C-band	5.5 cm	Surface deformation
Sentinel-2	SWIR	0.8-2.2 μm	Surface thermal anomaly
Sentinel-3	TIR	11-12 μm	Ash plume & Surface thermal anomaly
Sentinel-5P	UV	308-325 nm	Gas plume

Table 1. Sentinel Comparison for parameters to be monitored.

The MODIS operates both on Terra and Aqua spacecrafts with thirty-six spectral bands and three spatial resolutions at a temporal resolution of about 1-2 days. However, the spatial resolutions are 250m, 500m, 1000m which makes it less preferable where very high spatial resolution is a prerequisite. If the spatial resolution can be compromised, MODIS provides a much frequent and a large amount of data for use. Therefore, the choice of Sentinel-2 vs MODIS is dependent on the application requirements.

The Landsat8 operates at spatial resolutions of 30 meters (visible, NIR, SWIR), 100 meters (thermal) and 15 meters (panchromatic). It is the most continuous and available EO data

source with TIR channels acquired at medium spatial resolution (100 m). But the satellite's revisit frequency period of sixteen days imposes a limitation to measure volcanic activity with a suffice temporal resolution. Classification of all Sentinel-2 bands, Sentinel-2 bands that are common with Landsat-8 produced an overall accuracy of 5% and is 4% better than Landsat-8.

2.3 Sentinel-2 MSI Satellite

The Sentinel-2 Multispectral Instrument (MSI) satellite, a pivotal component of the European Space Agency's Copernicus program, stands at the forefront of Earth observation technology, providing essential data for a multitude of scientific endeavors. Its contribution to volcanology has been transformative due to its advanced features and capabilities.

Sentinel-2 MSI boasts a comprehensive array of 13 spectral bands, spanning from visible to shortwave infrared wavelengths. This diversity allows for the detailed analysis of various surface features, making it invaluable for discerning geological and environmental changes, including those associated with volcanic activities. The satellite's high spatial resolution, ranging from 10 to 20 meters, ensures precise and detailed imaging, capturing even subtle alterations on the Earth's surface.

In the realm of volcanology, Sentinel-2 MSI's capabilities are wide-ranging. Its spectral bands enable the identification of specific volcanic features, including lava flows, ash clouds, and gas emissions. The satellite's rapid revisit time of 5 days ensures timely monitoring of dynamic volcanic processes, aiding scientists in tracking changes over short periods, crucial for assessing potential hazards.

The availability of Sentinel-2 MSI data has democratized scientific research, allowing scientists globally to access high-quality, reliable information for their investigations. Its role in volcanology ranges from monitoring active volcanic sites to studying post-eruption impacts. By providing consistent, detailed, and timely data, Sentinel-2 MSI enhances our understanding of volcanic processes, aids in hazard assessment, and contributes significantly to the safety and well-being of communities residing in volcanic regions.

Therefore, the Sentinel-2 MSI satellite, with its advanced technology and freely accessible data, stands as an indispensable tool in volcanological research. Its impact is not only felt in the realm of scientific discovery but also in the practical applications related to disaster management and environmental protection.

2.4 Sentinel-2 MSI Data Overview

Sentinel-2 MSI captures data across **13 spectral bands**, ranging from visible and near-infrared to shortwave infrared. These bands enable detailed analysis of surface features, vegetation, and geological phenomena, crucial for volcanic studies. Each band has a specific wavelength range, allowing scientists to discern diverse materials and changes in the landscape.

Sentinel-2 MSI offers high spatial resolution, varying from 10 to 20 meters, ensuring detailed imaging of the Earth's surface. The wide swath of 290 km allows for extensive coverage, making it suitable for large-scale studies. This combination of high resolution and wide swath is particularly valuable for monitoring volcanoes with diverse features and active areas.

One of the key advantages of Sentinel-2 MSI is its short revisit time. It captures data of the same area every 5 days, providing frequent updates. This rapid revisit time is invaluable for monitoring dynamic volcanic processes, including changes in lava flow patterns, volcanic gas emissions, and vegetation cover alterations related to volcanic activities.

Sentinel-2 MSI data is distributed in **Level-1C and Level-2A products**. Level-1C data are calibrated at the top of the atmosphere, allowing for various corrections and analyses. Level-2A data, further processed to remove atmospheric effects, provide bottom-of-atmosphere reflectance values. These products are accessible through platforms like the Copernicus Open Access Hub, facilitating easy access for researchers and scientists.

Sentinel-2 bands	Sentinel-2A		Sentinel-2B		Spatial resolution (m)
	Central wavelength (nm)	Bandwidth (nm)	Central wavelength (nm)	Bandwidth (nm)	
Band 1 – Coastal aerosol	442.7	21	442.2	21	60
Band 2 – Blue	492.4	66	492.1	66	10
Band 3 – Green	559.8	36	559.0	36	10
Band 4 – Red	664.6	31	664.9	31	10

Band 5 – Vegetation red edge	704.1	15	703.8	16	20
Band 6 – Vegetation red edge	740.5	15	739.1	15	20
Band 7 – Vegetation red edge	782.8	20	779.7	20	20
Band 8 – NIR	832.8	106	832.9	106	10
Band 8A–Narrow NIR	864.7	21	864.0	22	20
Band 9 – Water vapor	945.1	20	943.2	21	60
Band 10 – SWIR – Cirrus	1373.5	31	1376.9	30	60
Band 11 – SWIR	1613.7	91	1610.4	94	20
Band 12 – SWIR	2202.4	175	2185.7	185	20

Table 2. Spectral bands for the Sentinel-2 sensors

3 Data Acquisition and Processing

Data Acquisition and Processing is vital for filtering only the required data avoiding various discrepancies in terms of Image quality, cloud disturbances, and covering the required geometry or AOI. In most cases the Sentinel-2A and Sentinel-2B data tiles only cover a partial region which in some cases can be merged by mosaicking.

3.1 Data Acquisition strategy

The data acquisition strategy employed in this study is a meticulous process ensuring the systematic collection of Sentinel-2 MSI images for accurate analysis. This strategy involves several key steps:

Selection of Acquisition Dates: Careful consideration is given to the selection of specific dates, driven by factors such as historical eruption patterns, volcanic activity reports, and weather conditions. Optimal dates are chosen, favoring cloud-free or minimally cloud-covered periods to guarantee the clarity of acquired imagery.

Various platforms have been used during this process to identify specific dates with volatile eruptive activities, these platforms include Sentinel EO browser, Global Volcanism Program, Planet explorer and Mounts project

Data Retrieval: Sentinel-2 MSI data is procured from reputable sources like the Copernicus Open Access Hub and other authorized data providers. Level-1C or Level-2A products are specifically targeted due to their radiometric and geometric accuracy, keeping in mind the Cloud cover percentage specific to our AOI. These products form the foundation for subsequent analyses.

3.2 Pre-processing Techniques

To enhance the quality and accuracy of the acquired images, essential preprocessing steps are meticulously executed. Atmospheric correction techniques are applied to mitigate the impact of atmospheric interference. Geometric rectification and radiometric calibration processes are implemented to ensure precise alignment and consistency across the dataset. These preprocessing measures are imperative, laying the groundwork for in-depth analysis and interpretation.

The Pre-processing of Sentinel-2 data has been carried out using ESA SNAP (Sentinel Application Platform) tool and the steps included in doing so are as follows.

Step-1: Resampling- Resampling refers to the process of changing the pixel resolution or grid of an image. This operation is commonly performed to match the pixel size of one dataset with another, ensuring consistency for analysis or visualization purposes.

The Optimum choice in our case for resampling is 'By Reference band from source product', therefore we can choose any band with resolution of 10m (B2, B3, B4, B8) and all the other bands will be resampled to match its size and resolution.

Resampling on Pyramid level is enabled as Pyramids are reduced-resolution representations of the dataset and are used to improve performance. Pyramids can speed up display of raster data by retrieving only the data at a specified resolution required for the display. Pyramids are created by resampling the original data. The resampling methods instruct the server how to resample the data to build the pyramids.

Step-2: Subset- a subset refers to a specific portion or region extracted from a larger satellite image or dataset. Subsetting is a fundamental operation in remote sensing and image processing, allowing users to focus their analysis on a particular area of interest within a

larger dataset. The subset (AOI) can either be chosen by specifying Pixel coordinates or Geographic coordinates by providing WKT for geometry.

It is optional to choose source bands based on the requirements or if no bands selected all the bands will be automatically processed. In Volcano lava flow detection as we know we study the SWIR bands that are B11, B12 and for information of the natural colors B2, B3, B4, B8 can be selected.

In addition to these processes, Mosaicking is done in some scenarios in order to merge two different Satellite images that are in adjacent tiles to recover our AOI. – Mosaicking is very useful when we have our area of interest in different tiles or at the end of the tile due to which it is not feasible to subset the AOI. Mosaicking is usually done for data captured at the same time in SWC and SWD, where the two products from different tiles can be combined into a single image for processing.

4 Extraction for Lava Flow mapping

Various methods can be employed for lava flow mapping, such as supervised or unsupervised classification algorithms. In a supervised approach, machine learning models are trained using labeled samples of lava flows and non-lava areas. Algorithms, such as Random Forest or Support Vector Machines, can be applied within SNAP to classify pixels into lava and non-lava classes. Unsupervised methods, like K-means clustering, can group pixels based on spectral similarities, enabling the identification of distinct lava flow boundaries without prior training.

4.1 Detection of Thermal Anomalies

In the context of lava flow mapping, identifying thermal anomalies is crucial as active lava emits significant heat that can be detected by thermal sensors on satellite imagery. ESA SNAP (Sentinel Application Platform) provides a powerful toolset for this purpose. First, the pre-processed Sentinel-2 satellite data is loaded into SNAP and is subjected to Spectrum Analysis to fully understand the thermal bands, sensitive to temperature variations. As discussed previously, our approach mainly focuses on the variations in spectral plots of Short wave Infrared (SWIR), Narrow Near Infrared and Near Infrared (NIR) bands by pointing at the lava flow regions. By applying appropriate band combinations and thresholding techniques, thermal anomalies indicating active lava can be differentiated from the surrounding terrain. This method leverages the thermal properties of the lava to identify its presence and activity.

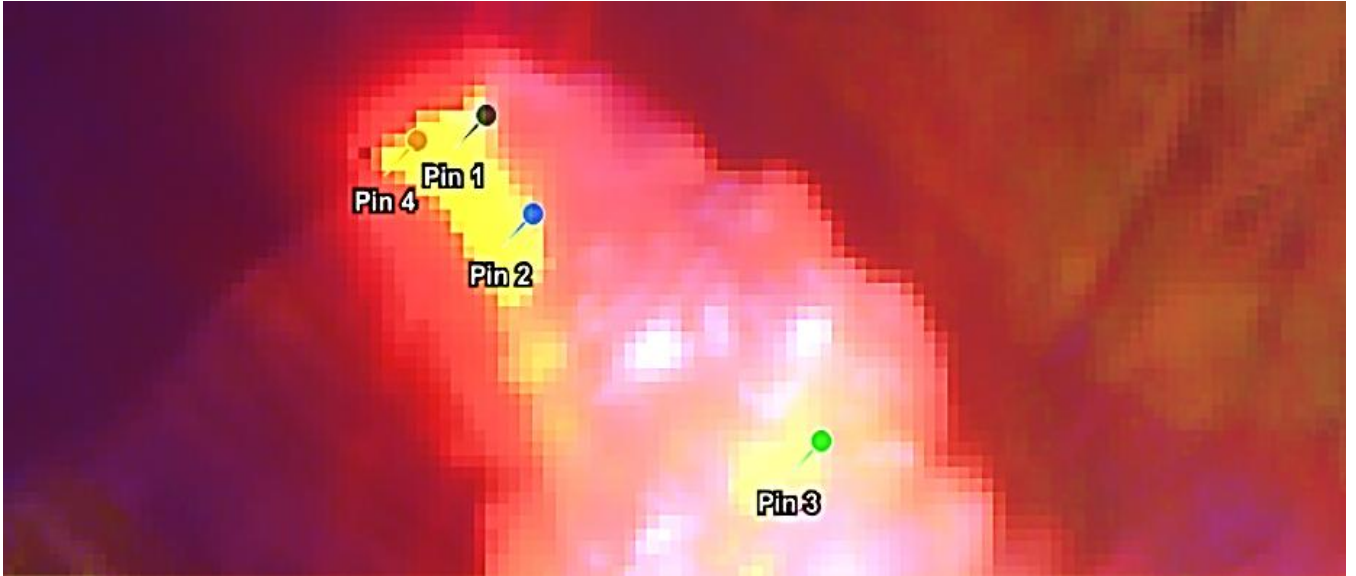


Figure 1. High Intensity Lava flow points

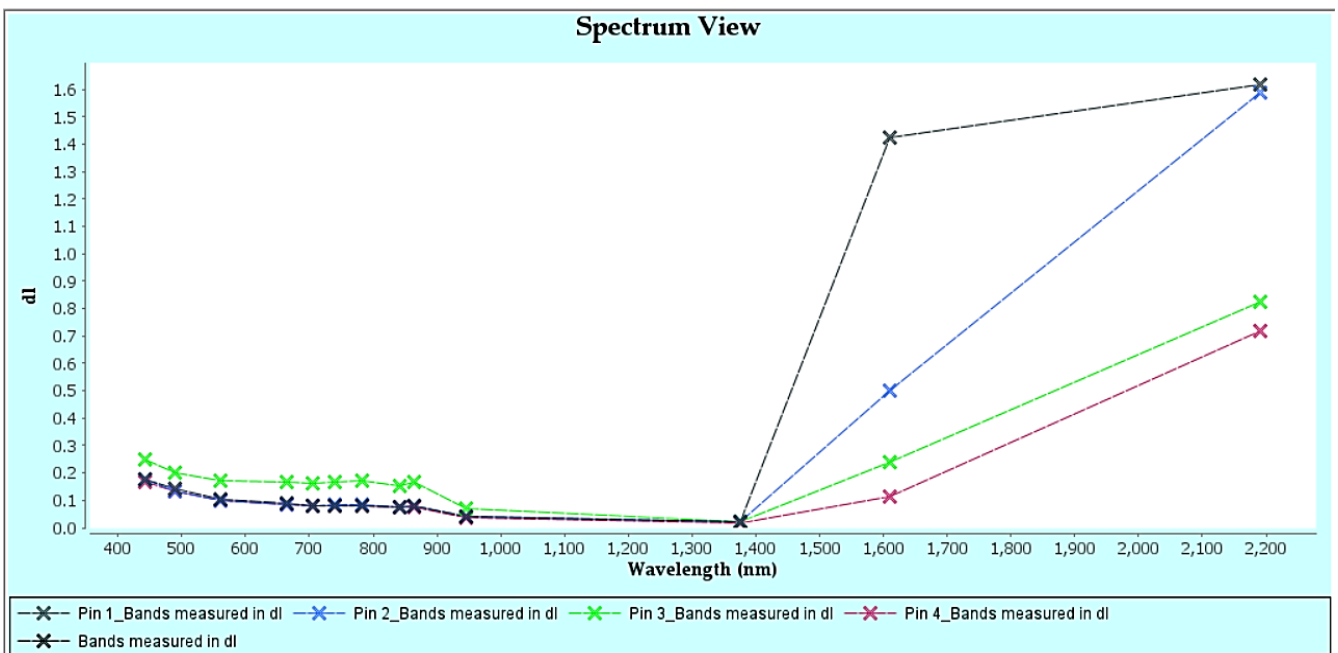


Figure 2. Spectrum view of High Intensity Lava flows

As can be inferred from the above Fig. 1 & 2, it can be noticed that for the pinned points on the lava flow, the intensity 'dl' of the SWIR and NIR bands tends to increase steeply. It is also seen that the band B12 is always greater than the threshold of 0.5 while the RGB bands B2, B3 and B4 are limited to less than 0.25. This exact behaviour has been constantly observed in a adequately large dataset of Sentinel-2 imagery ranging from 2017 to current date.

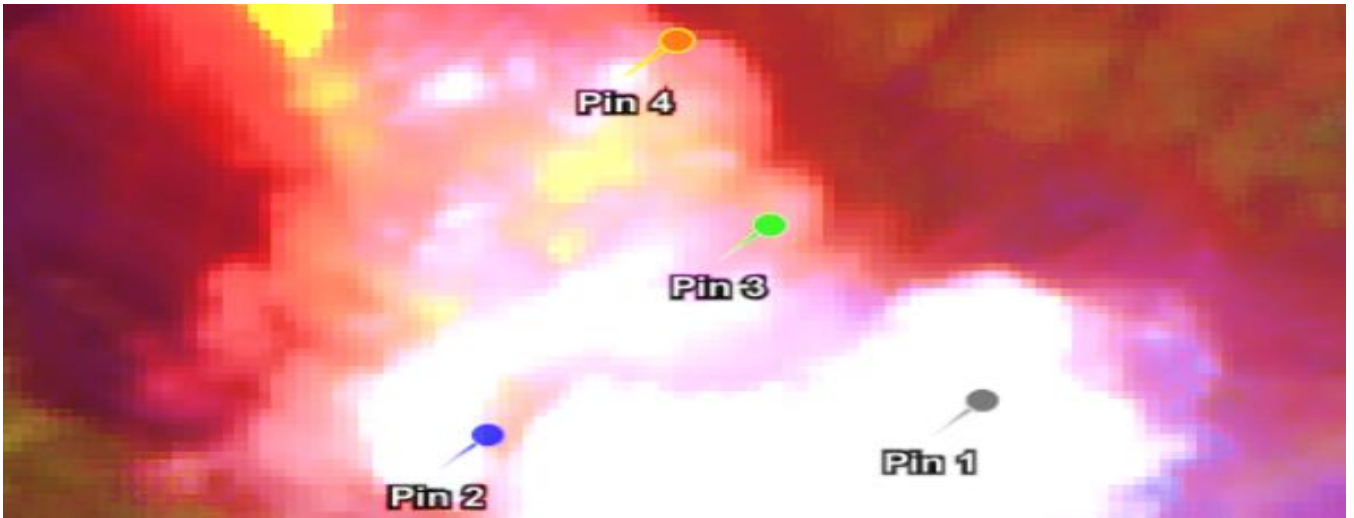


Figure 3. Satellite Imagery disrupted with Clouds.

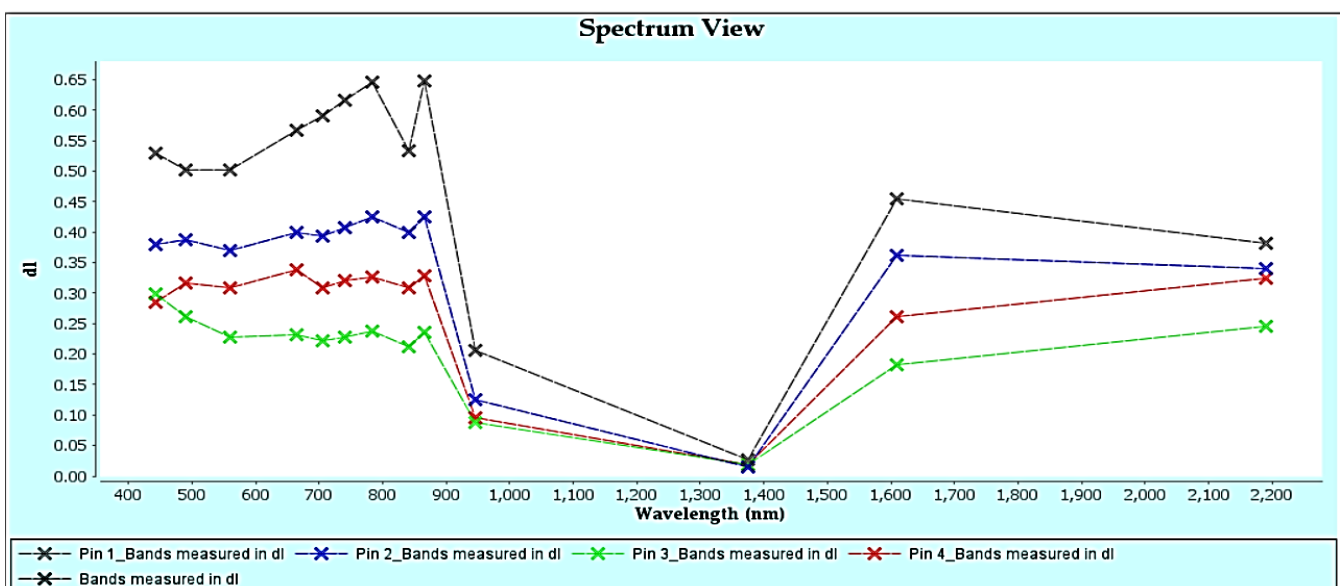


Figure 4. Spectrum view of Lava with cloud Interference.

While the previous observations can provide a clear understanding of how the lava flow is in line with the band's spectral intensity, there is an important issue of Cloud interference that can lead to falsification of these observations. Hence, to understand how the limits bands are being manipulated by the presence of clouds. Fig.3 and 4 represent the Spectral view of clouded areas, from which can be inferred the presence of clouds hugely increases the intensity of RGB B2, B3, B4 bands beyond the threshold of 0.25 that was previously observed at nominal conditions.

4.2 Lava flow detection and Classification

Once thermal anomalies are identified, the next step involves lava flow detection and classification. Lava flow detection and classification are critical components of volcanic monitoring systems, providing crucial insights for hazard assessment and disaster management. In this study, we present an innovative approach to analyzing and differentiating between active lava hot spots and mid-intensity lava flows using advanced spectral analysis techniques and Bandmaths formulations.

Spectral Analysis:

Spectral bands, including Short wave Infrared (SWIR), Narrow Near Infrared, and Near Infrared (NIR), are analyzed to identify variations indicative of thermal anomalies.

Detailed spectral analysis reveals distinct patterns in active lava regions compared to the surrounding terrain.

Bandmaths Formulations:

a. Hot Spot Detection (Bandmaths 1):

$$\text{Formulation: Hot Spot} = \frac{B12}{B8A}$$

Where the expression can be written as:

```
IF (B1 < 0.2 && B2 < 0.2 && B8A < 0.2) THEN
  (IF (B11 > 0.3 && B12 > 0.35) THEN 1
   ELSE (IF (B1 < 0.15 && B8A < 0.15) THEN
    (IF B11 > 0.2 && B12 > 0.275 THEN 1 ELSE NaN)
    ELSE NaN))
  ELSE (IF B12 > 1.75 THEN 1 ELSE NaN)
```

This formulation defines a conditional logic for detecting specific features or conditions in Sentinel-2 spectral bands. The algorithm evaluates various band intensities and thresholds to determine whether a pixel meets the specified criteria. If the conditions are met, a value of 1 is assigned; otherwise, the pixel value is set to NaN (Not a Number).

The HotSpot index is calculated to identify intense thermal activity, characterizing active lava hot spots. Higher index values indicate areas with significant thermal emissions.

b. Mid-Intensity Lava Detection (Bandmaths 2):

$$\text{Formulation: Mid-Intensity Lava} = \frac{B8A - B11}{B8A + B11}$$

Where the expression can be written as:

```
IF (B1 < 0.2 && B2 < 0.2 && B8A < 0.2) THEN
  (IF (B11 > 0.2 && B12 > 0.3) THEN 1
   ELSE (IF (B1 < 0.15 && B8 < 0.15) THEN
    (IF B12 > 0.275 THEN 1 ELSE NaN)
    ELSE NaN))
  ELSE (IF B12 > 1.75 THEN 1 ELSE NaN)
```

The Mid-Intensity Lava index is computed to detect mid-intensity thermal activity, distinguishing slower-moving or cooling lava flows. Positive index values indicate the presence of mid-intensity lava.

During the process of Bandmath Calculation various formulations has been identified, however each one seems to have a certain restriction and ambiguity, a trial-and-error formulations with their limitations are indicated as below:

IF (B1<0.2 && B2<0.2 && B8A<0.2) THEN (IF (B11>0.3 && B12>0.35) THEN 1 ELSE (IF (B1<0.15 && B8A<0.15) THEN (IF B12>0.75 THEN 1 ELSE NaN) ELSE NaN)) ELSE (IF B12>1.75 THEN 1 ELSE NaN)

The above formulation has been implied for the Hot-Spot detection, but in doing so a few pixels of hotspots which are in verge of starting to cooldown have been neglected.

IF (B1<0.25 && B2<0.25 && B8A<0.25) THEN (IF (B11>0.16 && B12>0.5) THEN 1 ELSE (IF (B1<0.2 && B8A<0.18) THEN (IF B12>0.37 THEN 1 ELSE NaN) ELSE NaN)) ELSE (IF B12>1.75 THEN 1 ELSE NaN)

This expression is subjected to Mid-Intensity lava flow detection and seem to have a similar limitation as the above and also picks up a few pixels relating to the clouds of L1C sentinel-2 Images, therefore leading to a process of Cloud masking using IDEpix for Sentinel-2.

4.3 Validation and Accuracy assessment

Validation and accuracy assessment are integral parts of any remote sensing analysis, ensuring the reliability and precision of the derived information. In the context of our lava flow detection and classification study using Sentinel-2 satellite data, it is crucial to validate the methods employed and assess their accuracy. This section outlines the validation procedures and accuracy assessment techniques employed to verify the results obtained from our analysis.

Upon generating the band masks using SNAP, the bands are exported as raster data on QGIS to compute the overall Lava flow areas using the raster layer unique values report. The obtained results from QGIS have been then correlated with a Google Earth Engine API 'NHI-tool' that uses Landsat 8/9 and Sentinel 2 collections for the investigation of volcanic thermal anomalies and compute the total Lava flow areas for the purpose of validation.

The changes in the outcome of Lava flow areas and accuracy in the project work compared to the ML algorithm of NHI-tool have been speculated and reported in the *Tables.3, 4, 5, 6 & 7* found in Appendices.

The validation procedures and accuracy assessment techniques outlined in this section are essential for ensuring the credibility of our lava flow detection and classification results. These metrics provide quantitative measures of classification accuracy and the agreement between the classified results. The differences in computed outputs is presented in the form of plots, thus making it easier to predict the correlation between the obtained values and reference tool.

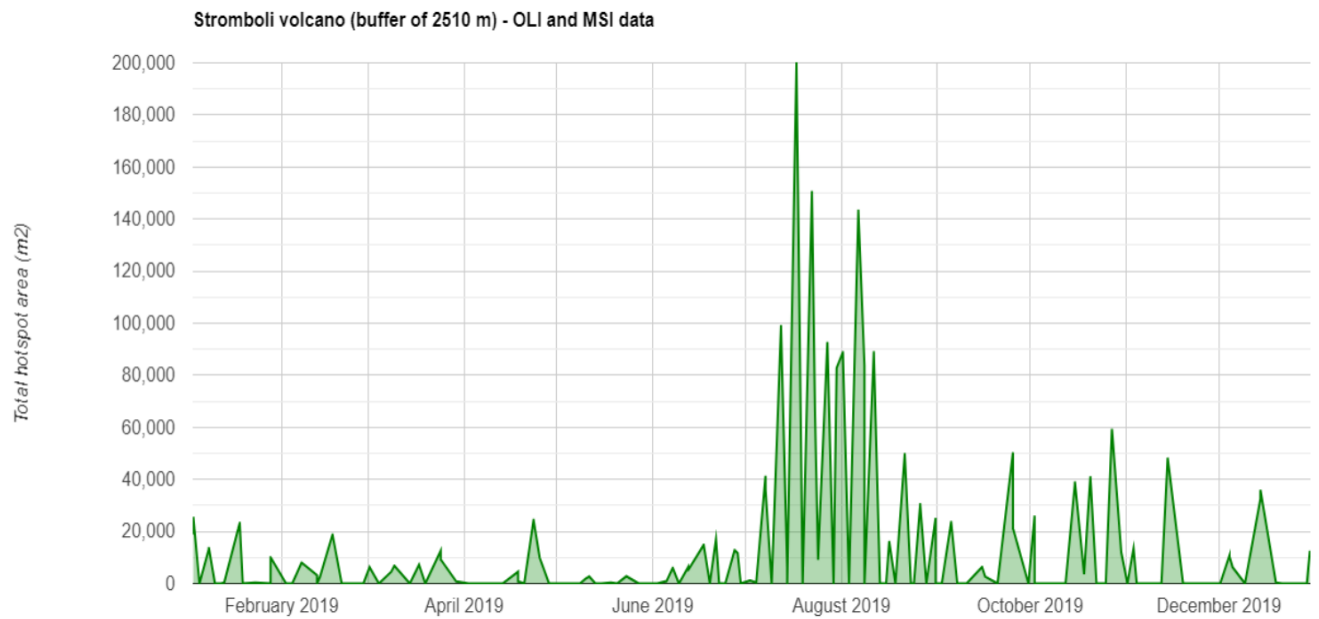


Figure 5. 2019 NHI- tool Total Hot-Spot Area in m2

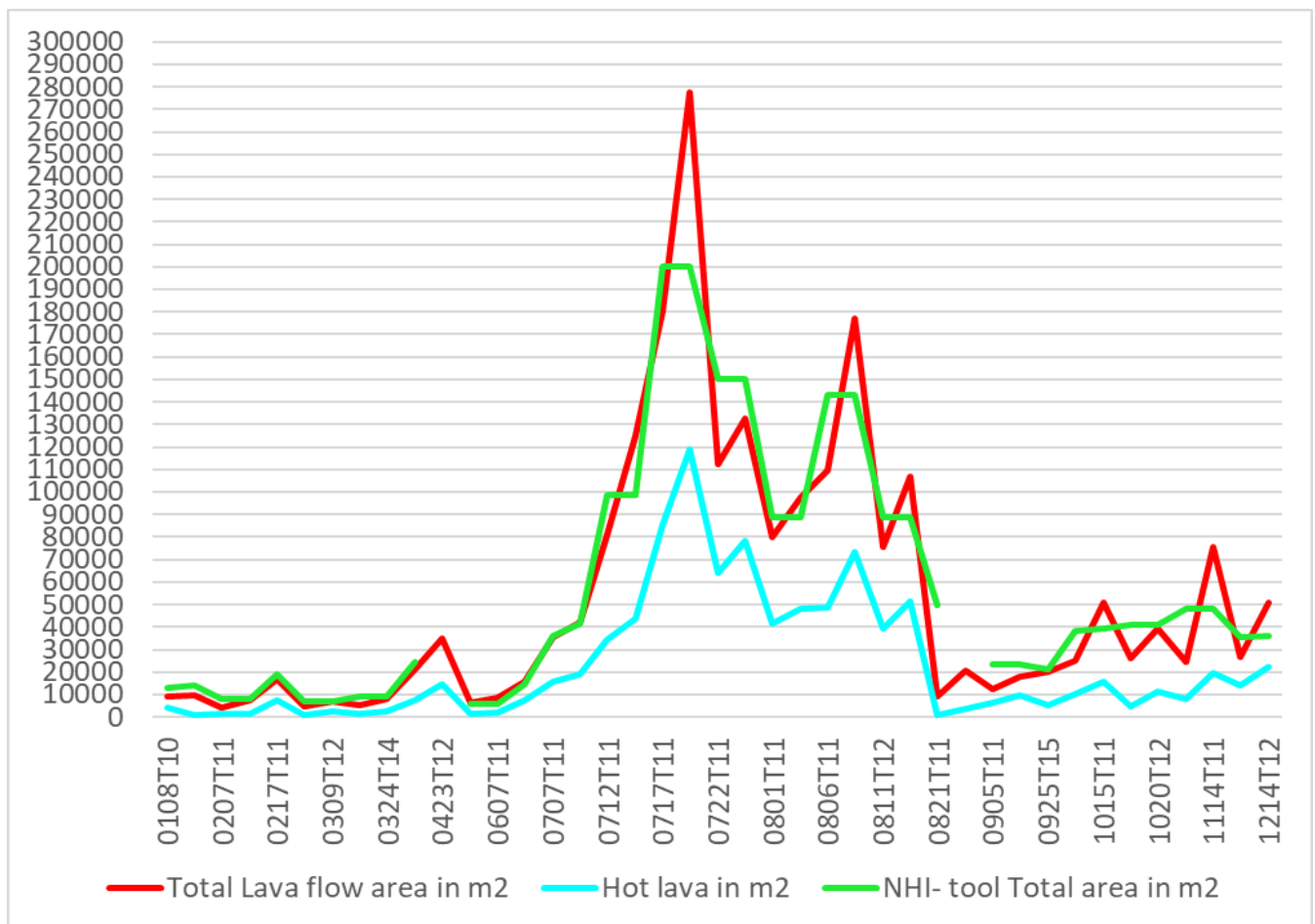


Figure 6. 2019 NHI-tool vs Thesis Total Lava flow area comparison

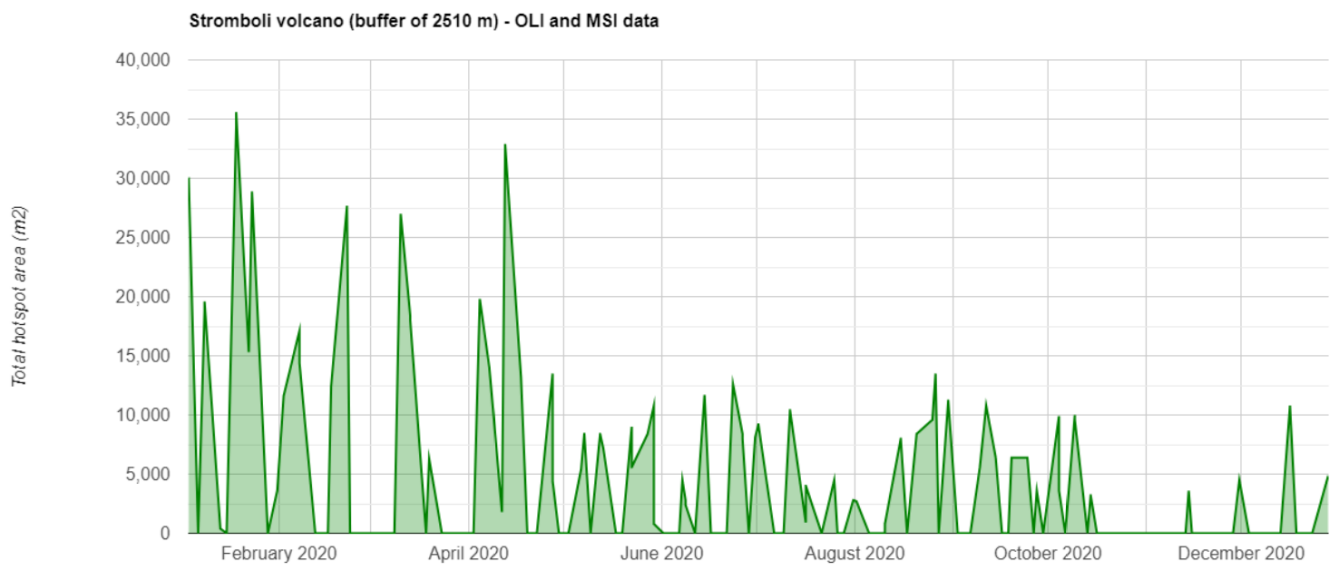


Figure 7. 2020 NHI- tool Total Hot-Spot Area in m2

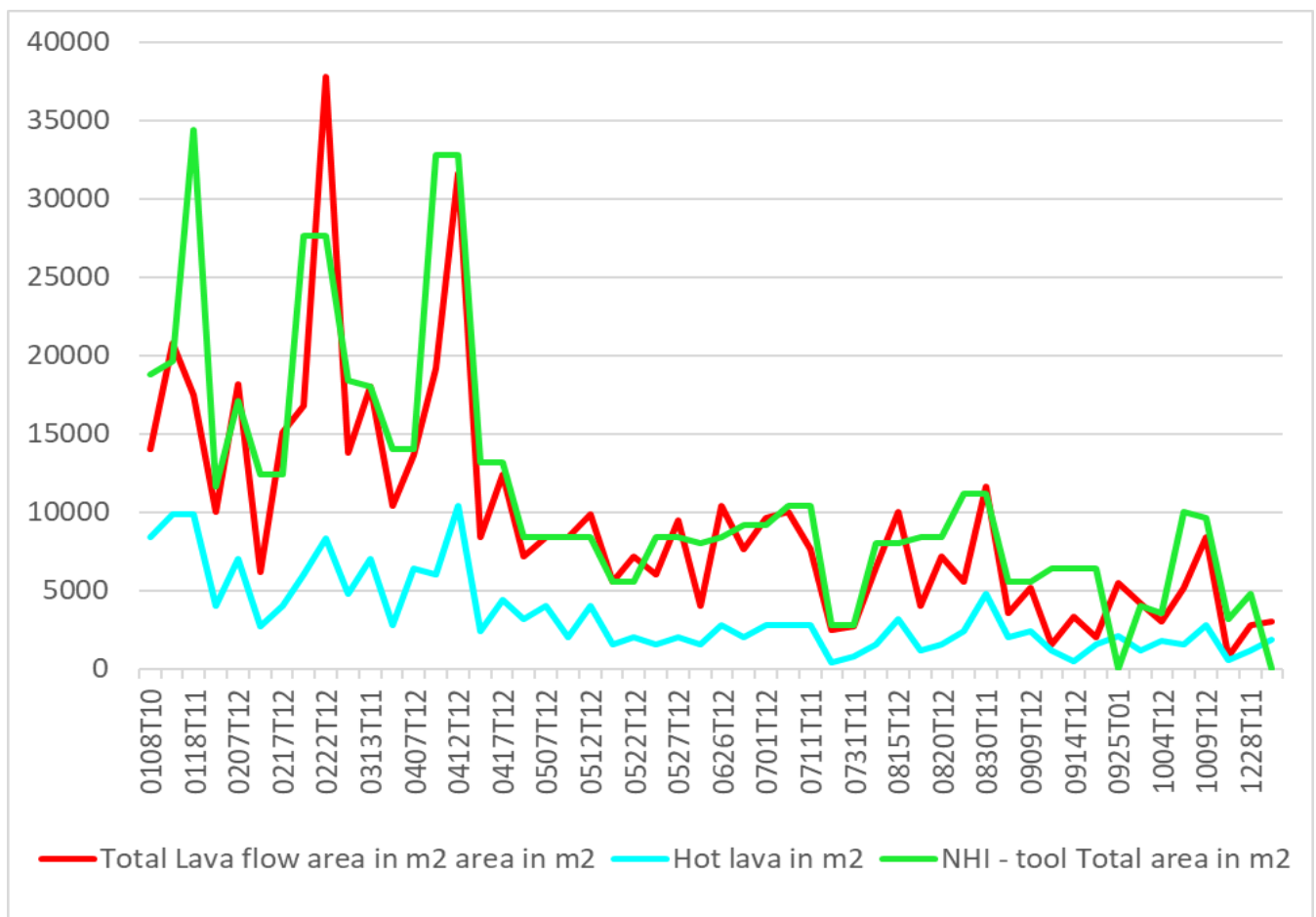


Figure 8. 2020 NHI-tool vs Thesis Total Lava flow area comparison

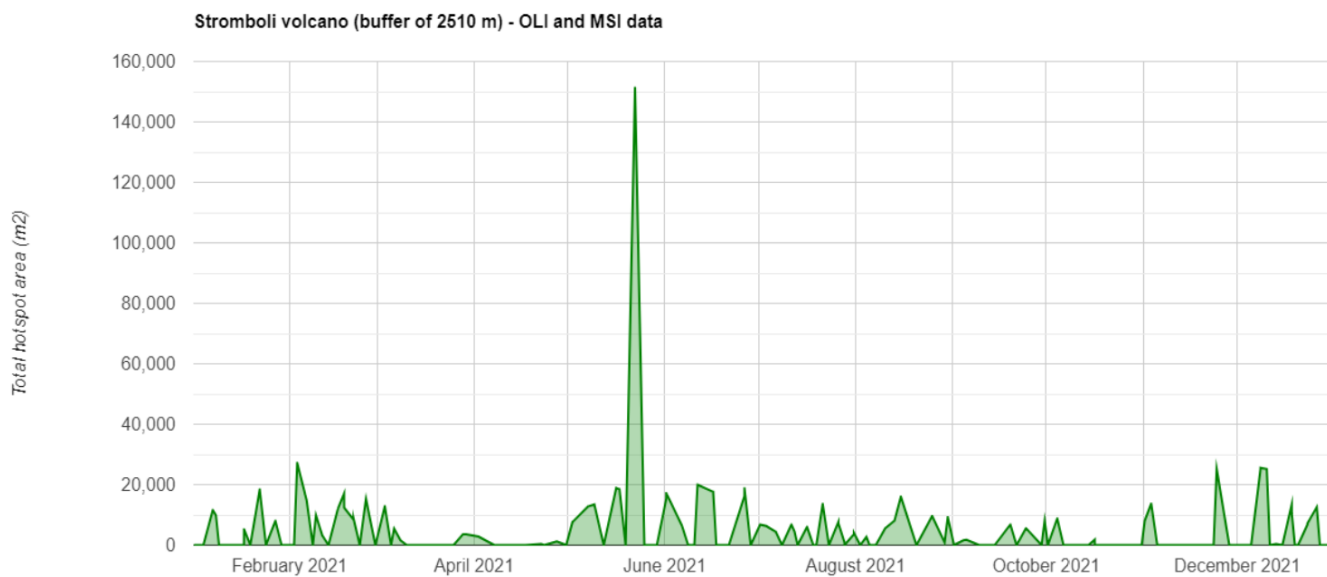


Figure 9. 2021 NHI- tool Total Hot-Spot Area in m2

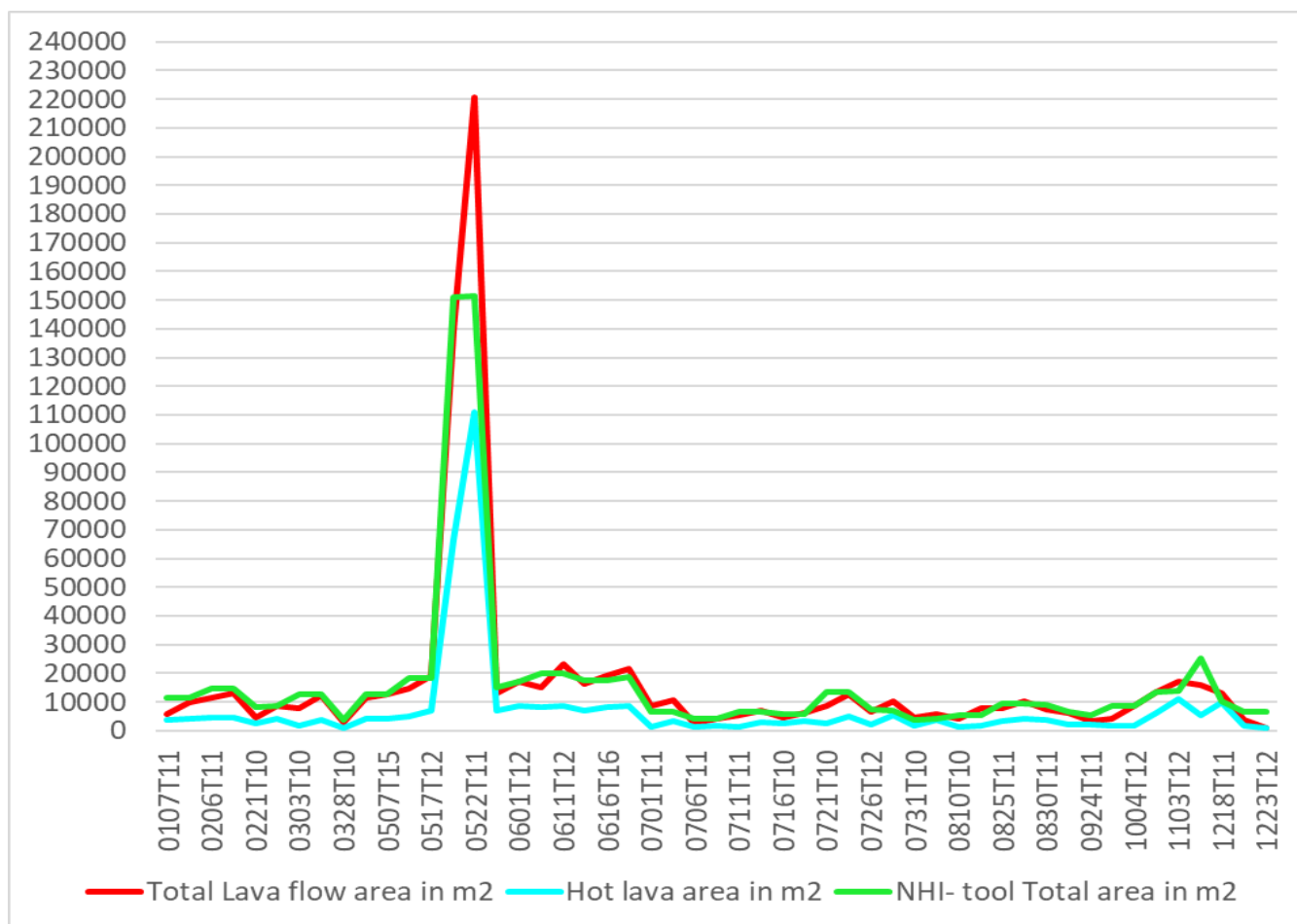


Figure 10. 2021 NHI-tool vs Thesis Total Lava flow area comparison

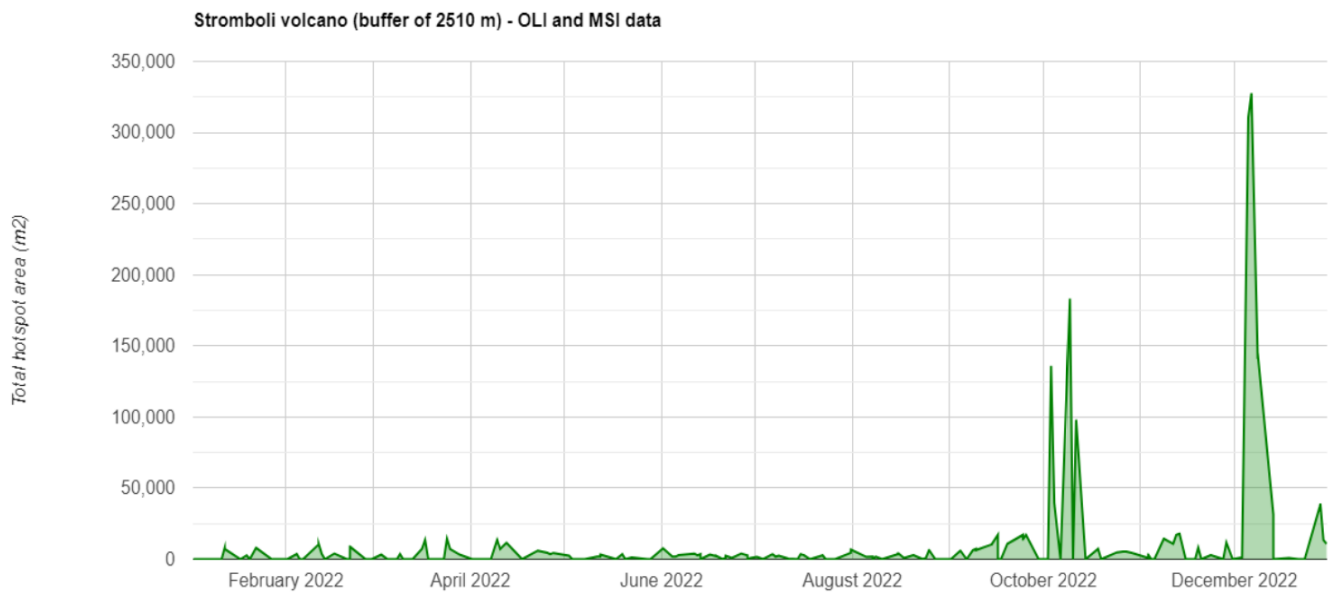


Figure 11. 2022 NHI- tool Total Hot-Spot Area in m2

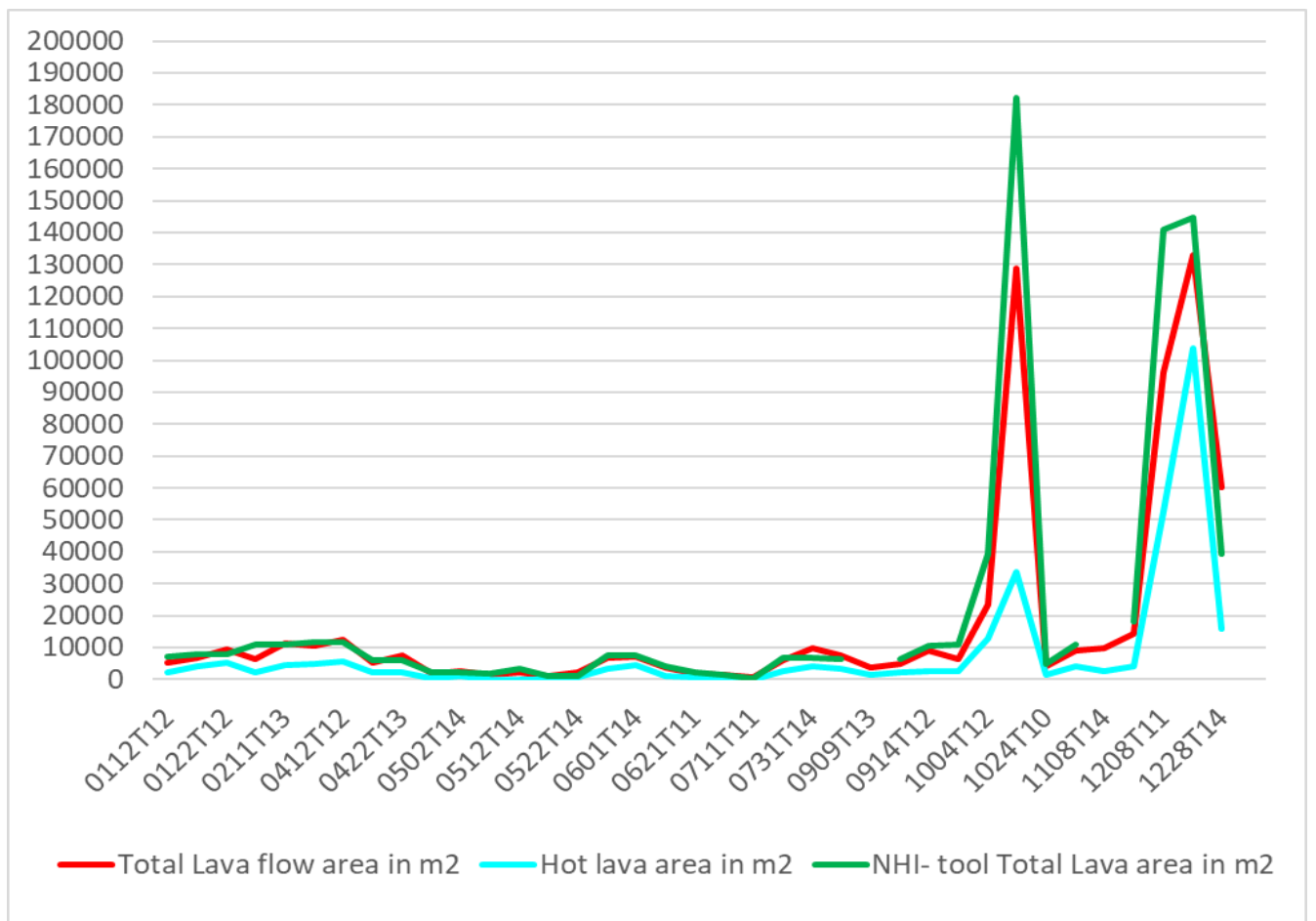


Figure 12. 2022 NHI-tool vs Thesis Total Lava flow area comparison

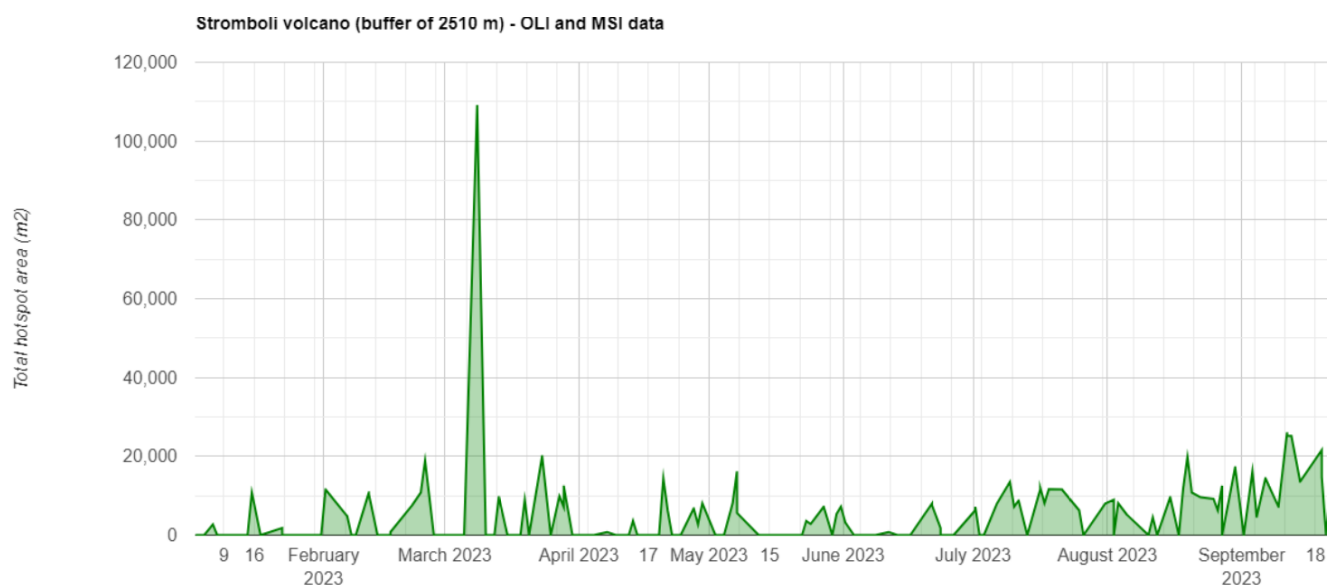


Figure 13. 2023 NHI- tool Total Hot-Spot Area in m2

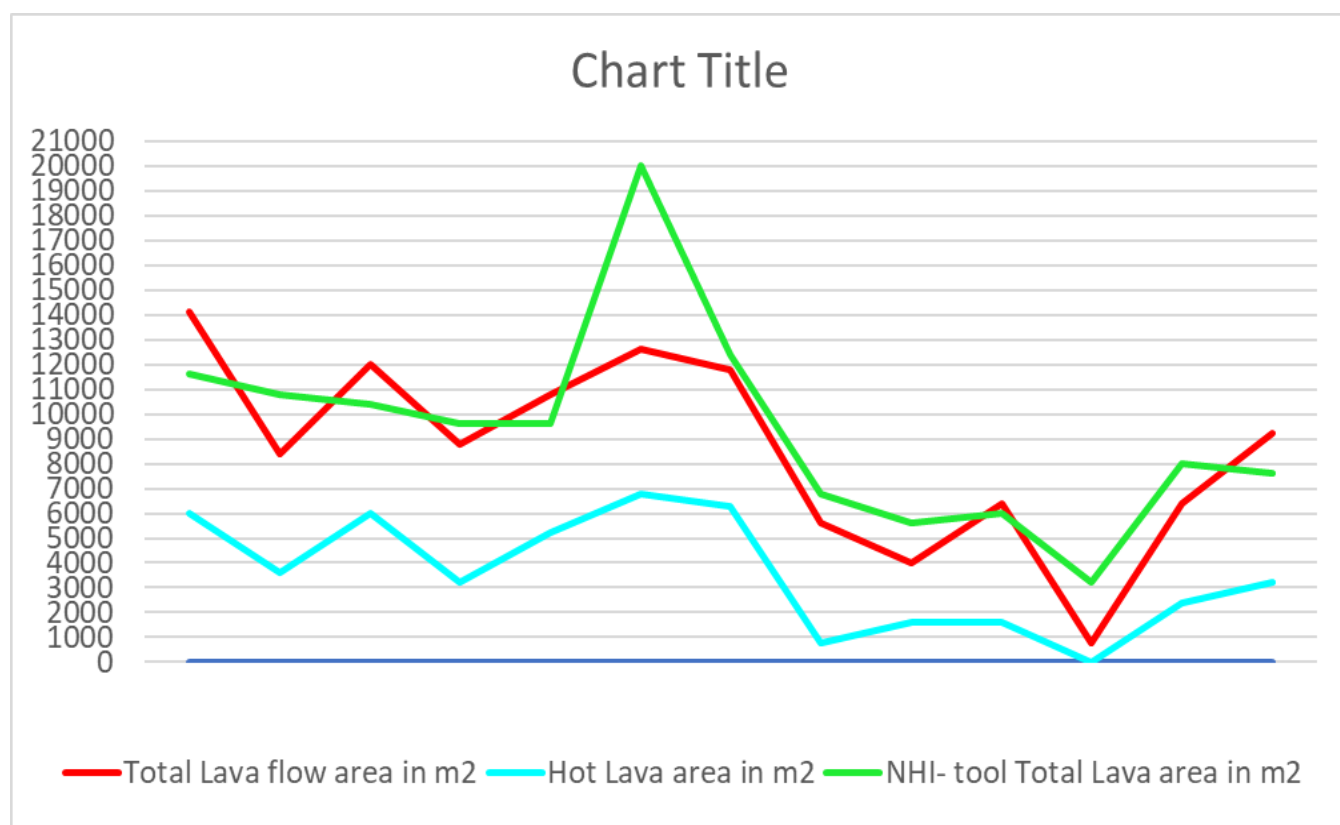


Figure 14. 2023 NHI-tool vs Thesis Total Lava flow area comparison

Our study aimed to detect and analyze lava flows on Stromboli using Sentinel-2 satellite imagery. We compared our results with those generated by the NHI tool, which utilizes both

Sentinel-2 and Landsat 8 data. The comparative analysis revealed several important points of consideration:

1. Temporal Resolution Discrepancy:

Thesis Work: Our study computed increasing lava flow between time intervals, providing a more detailed temporal analysis.

NHI Tool: The NHI tool computes or clubs all imageries into a single occurrence of lava for the day, potentially missing temporal nuances within shorter time intervals.

2. Resolution Variation:

Thesis Work: The use of Sentinel-2 data in our study maintained consistent pixel resolution, ensuring accuracy in lava flow delineation.

NHI Tool: The combination of Sentinel-2 and Landsat 8 data in the NHI tool introduces variation in pixel resolution, potentially affecting the precision of lava flow detection.

3. Consideration of Thermal Effects:

Thesis Work: Cloud interference was neglected in our thesis work, which could lead to inaccuracies in lava flow detection due to thermal effects of clouds and ash fumes.

NHI Tool: The NHI tool considers appropriate thermal effects of clouds and ash fumes, enhancing the accuracy of lava area detection.

4. Comparative Outcome:

The outcomes from our study and the NHI tool were approximately similar, indicating the effectiveness of both methods in detecting lava flows.

While both methods detected similar lava areas, the variation in pixel resolution accounted for the differences in the calculated area values.

5. Precision of Lava Area Detection:

Thesis Work: The images from our thesis work precisely identified the lava area, demonstrating the effectiveness of our methodology in this specific scenario.

NHI Tool: The resolution differences explained the variations in the detected area, but the lava flow patterns appeared consistent between the two methods.

4.4 Temporal Evolution of Lava flows

In this analysis, we delve into the temporal evolution of lava flows using Sentinel-2 satellite imagery, focusing on the dynamic changes in flow patterns, intensities, and geographical extents over distinct time intervals. We can proceed to observe how the changes takes place with the help of Processed Sentinel-2 imagery.



Figure 15. 2019, January and February Volcano Activity

Based on the processed Sentinel-2 satellite images in Fig.15 for the period between January and February 2019, it can be observed that Stromboli displayed a period of reduced activity during those months. The data from these images indicate a lower level of volcanic activity during that specific timeframe.



Figure 16. 2019, March and April Volcano Activity

The March of 2019 images still demonstrates a period of reduced activity, from April 2019 there is a notable increase in the lava emission.



Figure 17. 2019, June volcanic activity

Particularly in the month of June there is a gradual increase in the spitting of lava between a period of 10 days.

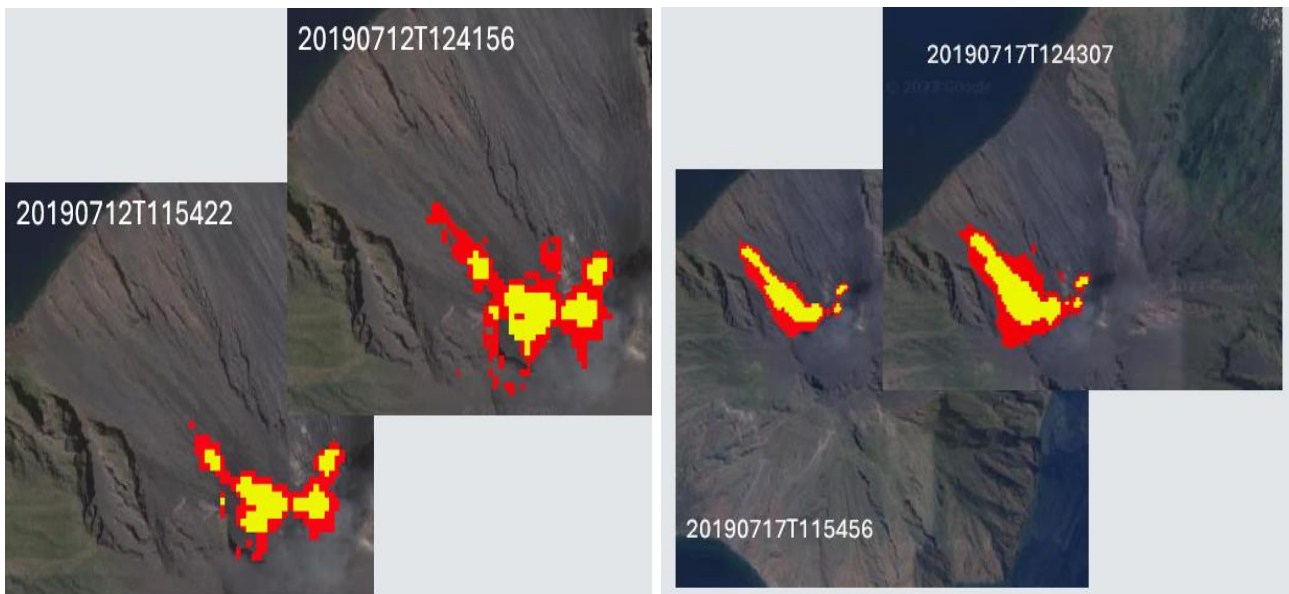


Figure 18. 2019, July volcanic activity

In July 2019, Stromboli experienced a period of intense activity, marked by substantial lava emissions. This heightened volcanic activity was clearly discernible in the Sentinel-2 satellite images, which were captured every five days. These images vividly captured the significant outpouring of molten lava from the volcano during that time.

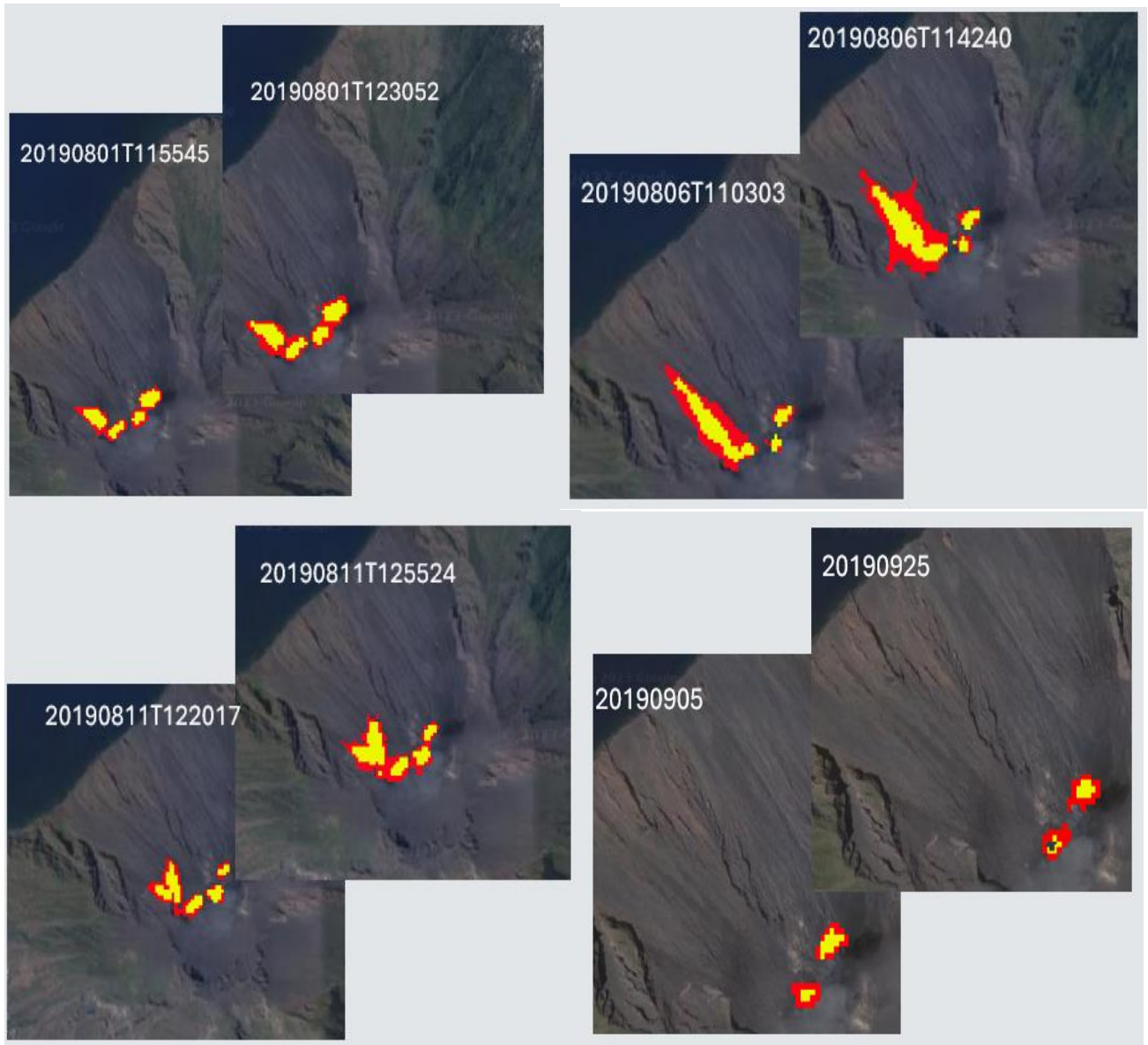


Figure 19. 2019, August Stromboli Lava flow

Following July 2019, In August 2019, Stromboli exhibited a period of heightened activity, featuring substantial lava emissions. This increase in volcanic activity was clearly observable in the Sentinel-2 satellite images, captured at five-day intervals. Specifically, the lava emissions noticeably escalated from August 1st to August 6th, 2019, before gradually decreasing between August 6th and August 11th, 2019 and eventually decreases during the month of September 2019. These fluctuations in lava emission were distinctly captured through the Sentinel-2 imagery during this timeframe.

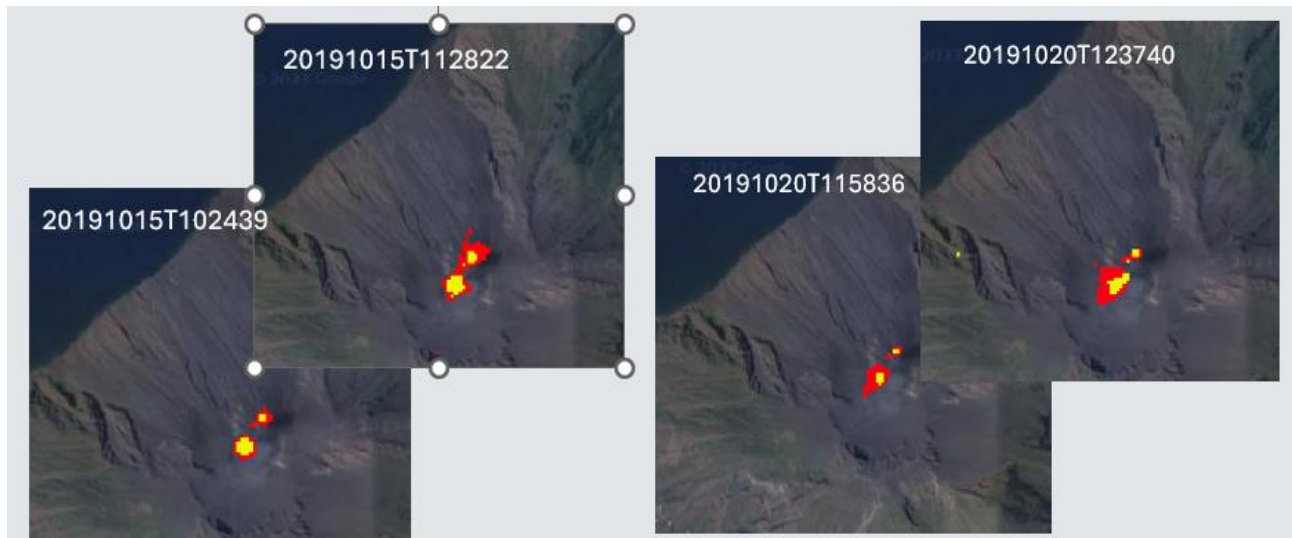


Figure 20. 2019, October Stromboli Lava flow

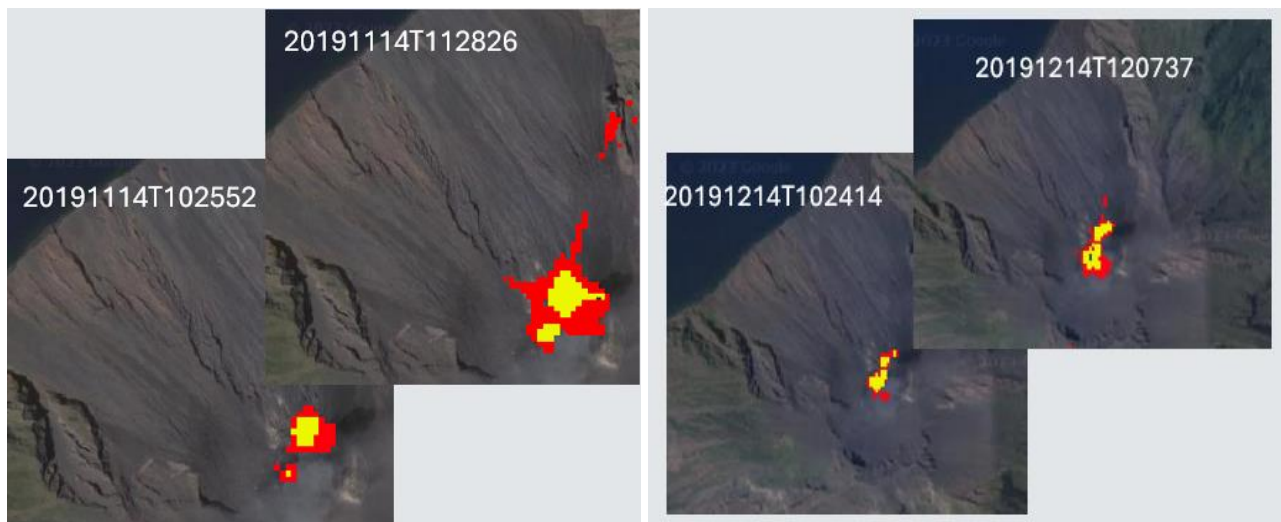


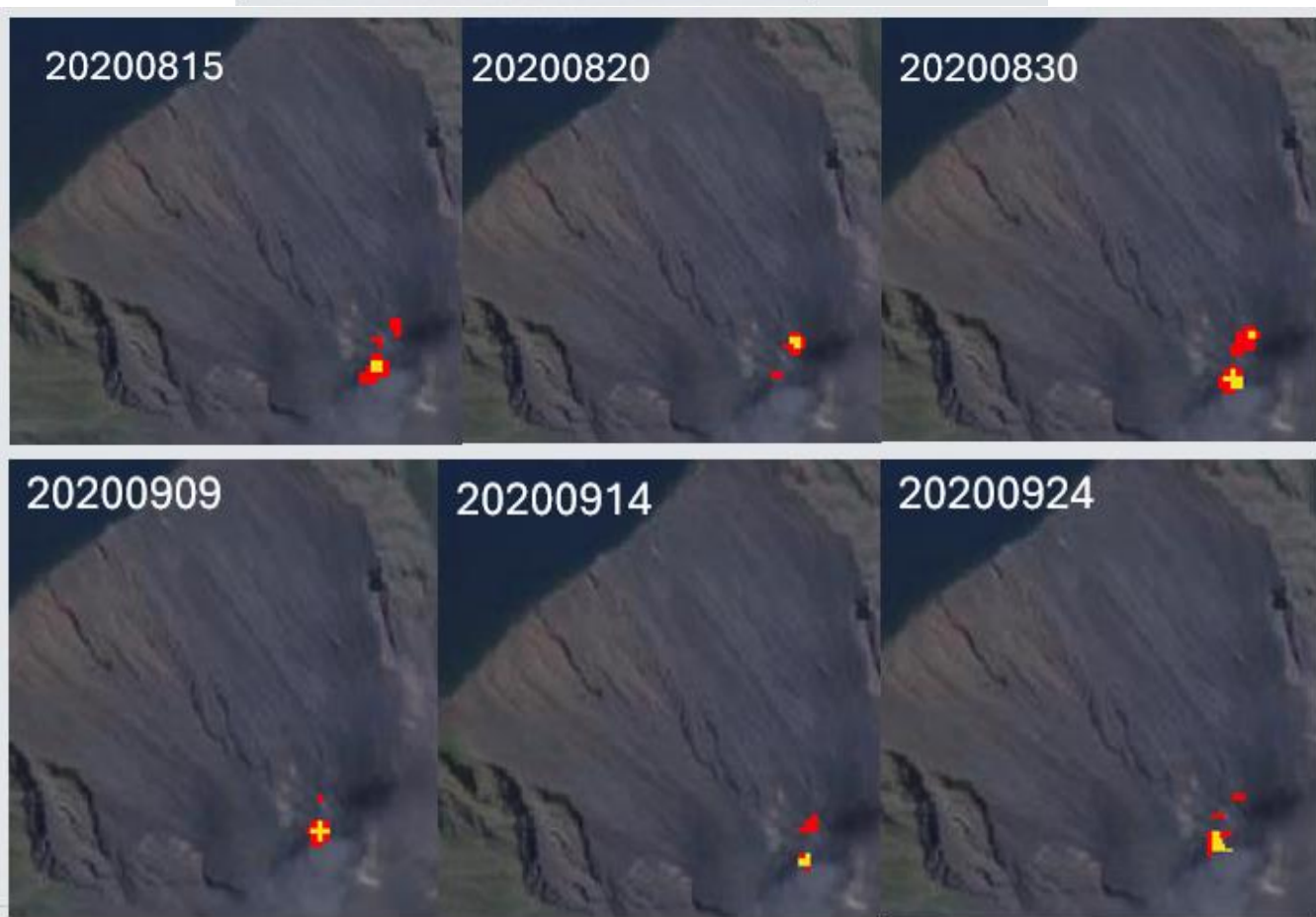
Figure 21. End of 2019 Stromboli activity

Based on the analysis of Sentinel-2 images taken during November and December 2019, the activity of Stromboli appeared to be in a state of neither activation nor deactivation. These images did not indicate any significant increase or decrease in volcanic activity during that period. Stromboli seemed to maintain a relatively stable condition, showing no noticeable signs of heightened eruptions or complete inactivity during those months.

2020









2021





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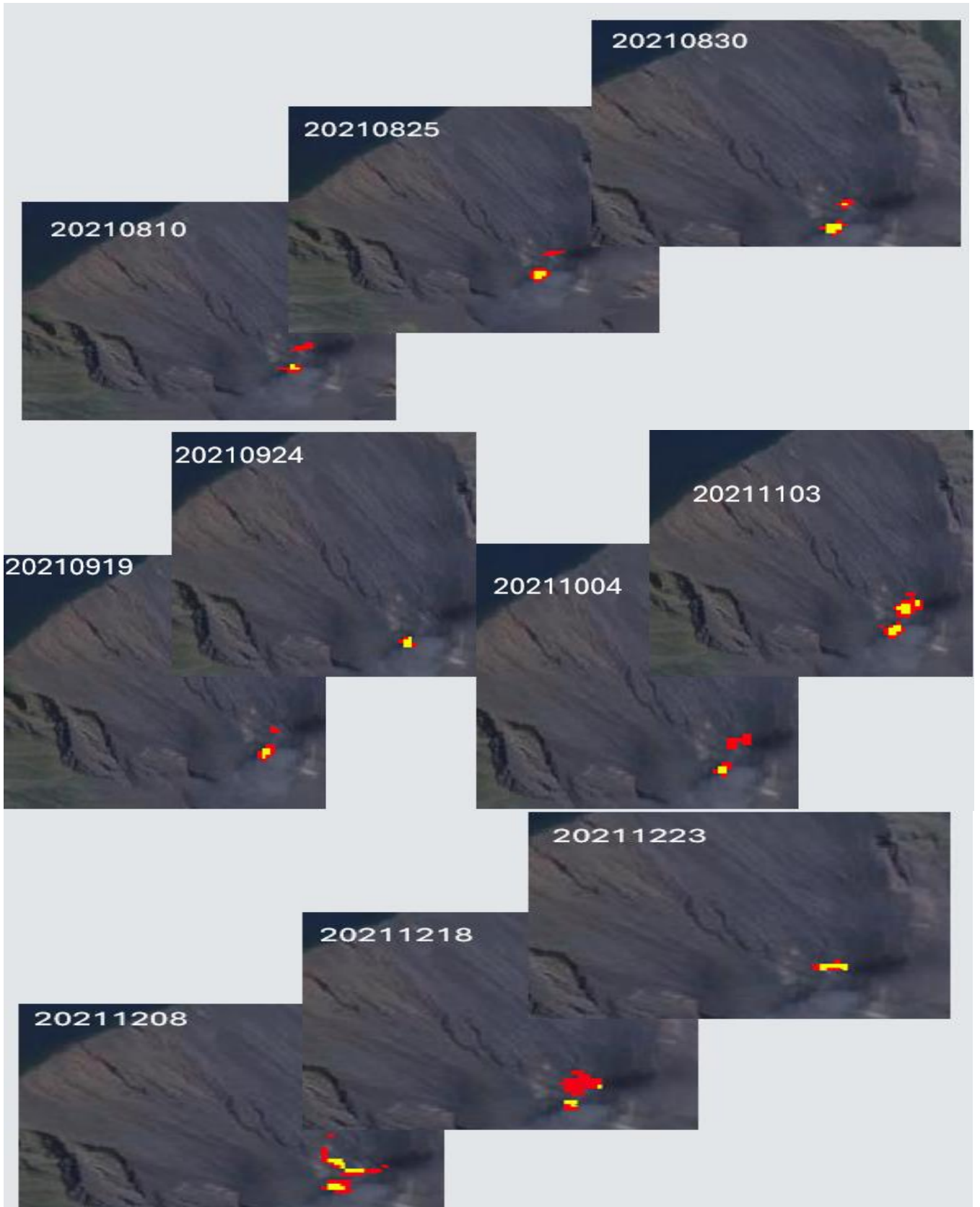
20210919

20211004

20211223

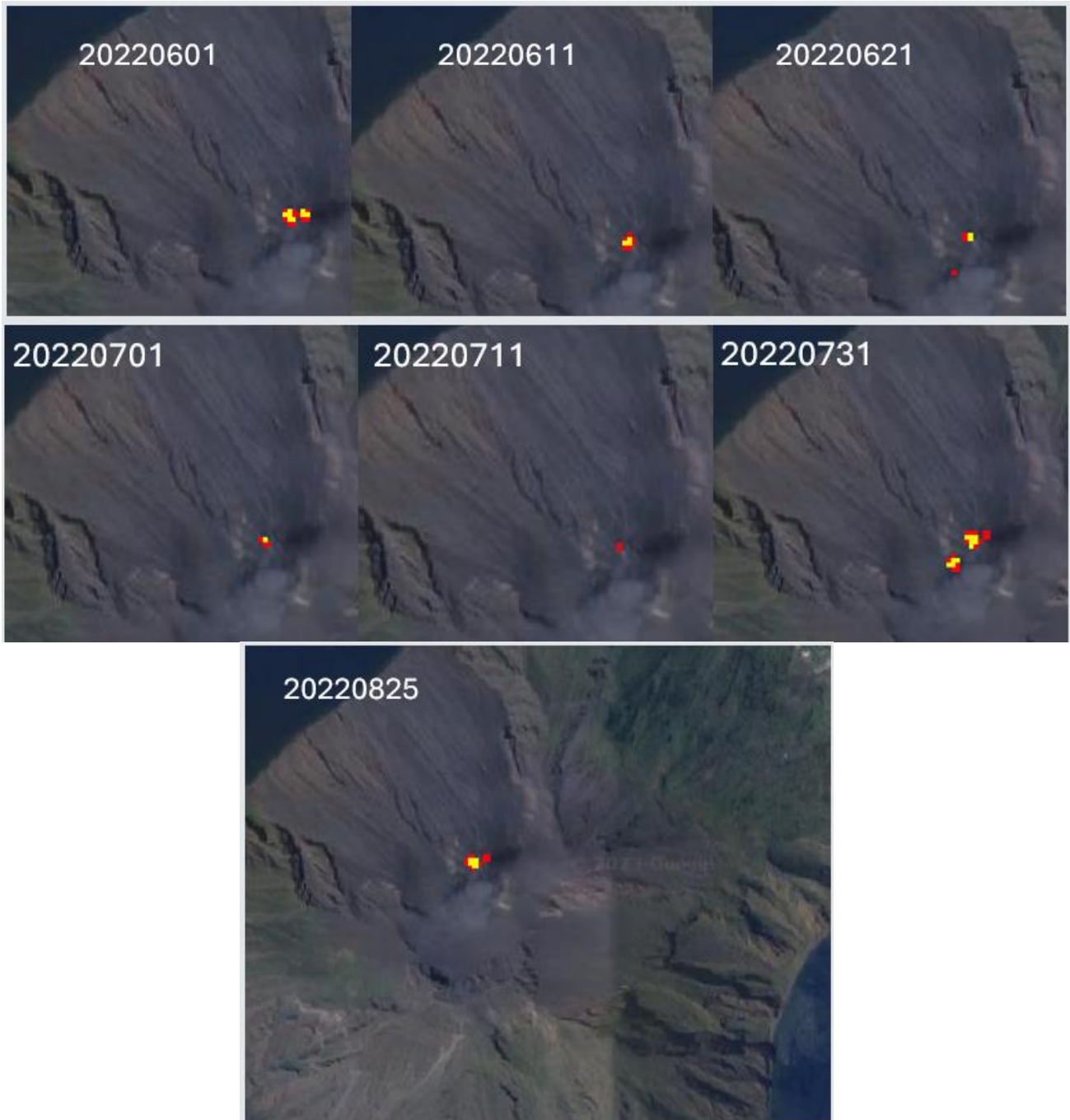
20211218

20211208



2022

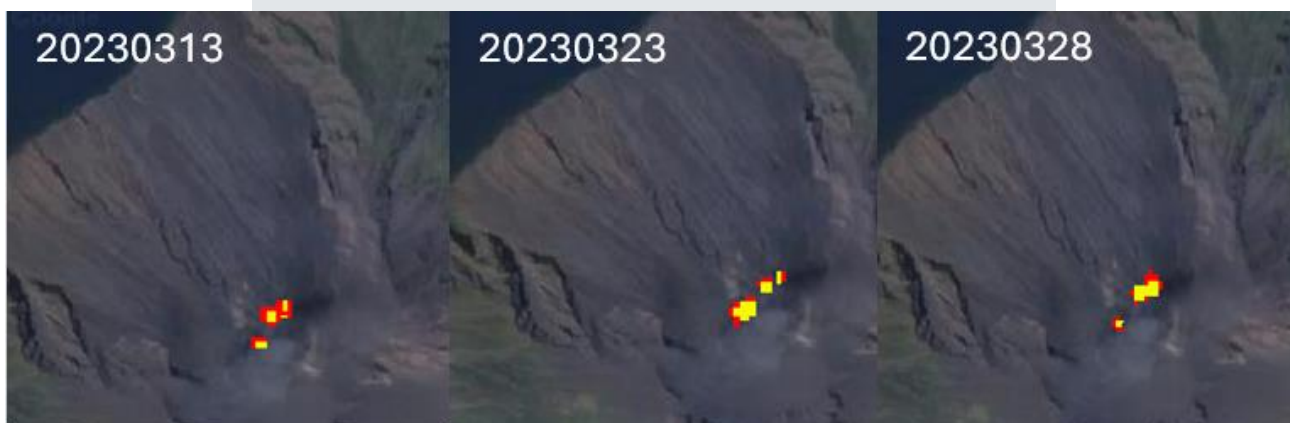
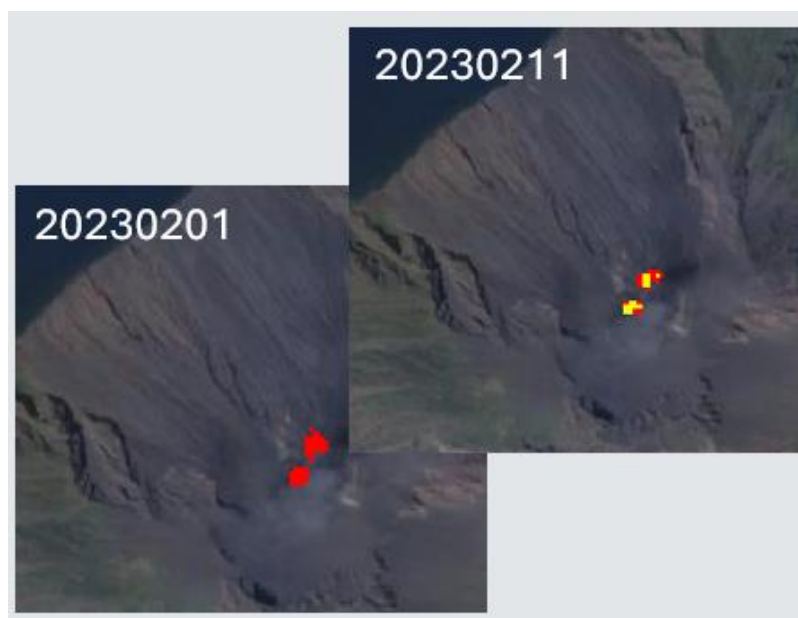








2023





These Sentinel-2 processed data are co-related with the consecutive acquisitions which helps us to understand how the Volcanic activity takes place. Although the patterns that can be observed here are precise, these data can be used along with the ground data for real time analysis as the Sentinel-2 acquisitions have longer time intervals that makes it an unsuitable option for real time case studies and monitoring.

4.5 Comparison with NHI-tool Lava Flows

As we have previously during the Validation of our computed lava flow with the GEE NHI-tool, it can be noticed that there were various reasons due to which there is a difference in the Area of lava flow obtained. In this chapter, we can discuss more in detail about these causes with the comparison of Processed remote Sensing data.

Statement – A: The tool has computed only a single Occurrence of Lava of the day whereas from the Sentinel-2 it can be seen there is an increasing Lava flow between time intervals.

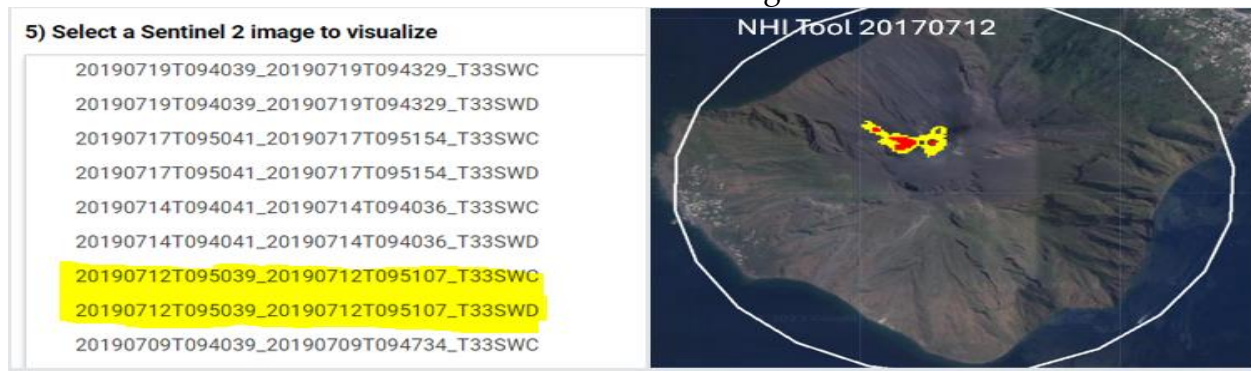


Figure 15. NHI Processing of one image 20190712



Figure 16. Thesis work, Sentinel-2 images processed for 20190712

Statement- B: The NHI tool uses Sentinel-2 and LandSat8 data for computation, which tends to variation in the resolution of pixels.

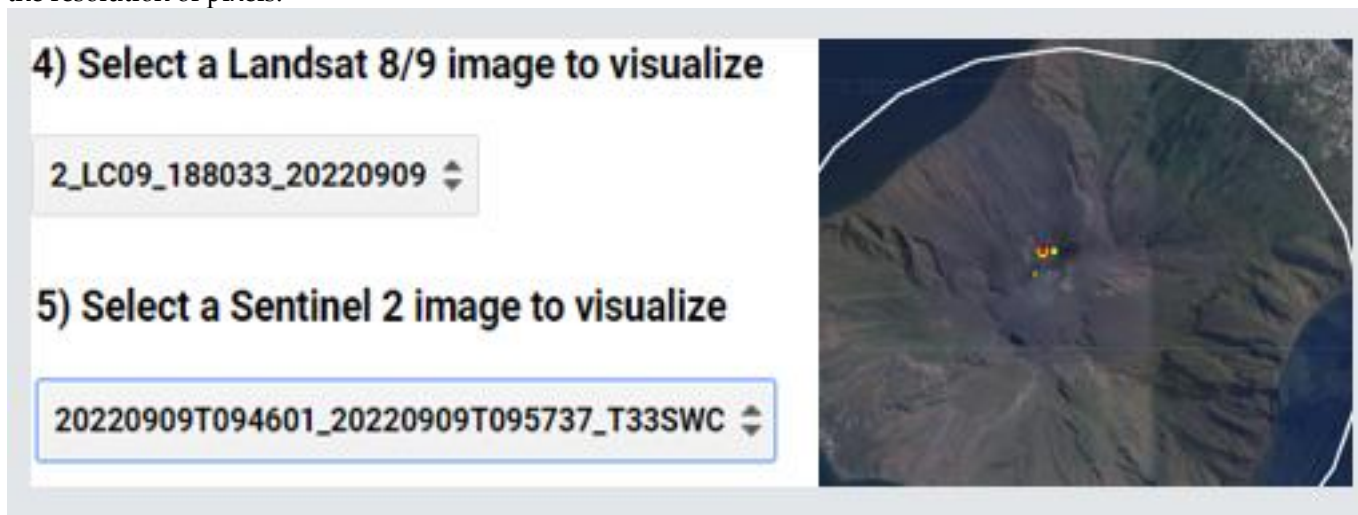


Figure 17. NHI-tool processing both Landsat and Sentinel Images for 20220909

Statement – C: The NHI algorithm detects the Lava area considering appropriate Thermal effects of Clouds, whereas the thesis work neglects Cloud Interference and Ash fumes.



Figure 18. Detection of Thermal reflectance behind the Ash Fumes by NHI-tool

Statement – E: Comparing the Images obtained from thesis work and NHI tool, Thesis work seems to precisely pick the lava area in this scenario, and only the resolution considered by NHI algorithm explains the difference in Area.



Figure 19. NHI vs Thesis Accuracy

Statement – F: The images do not seem to differ between the computed work and NHI tool, however there is a difference in area Value.



Figure 20. Difference in Area Value while the Hotspot computation doesnt differ

4.6 Environmental Impact Assessment

The environmental impact assessment using Sentinel-2 satellite data provides valuable insights into the ecological consequences of Stromboli's volcanic activities. By analyzing changes and deformation in land cover, lava flows, and gas emissions, authorities and researchers gain a comprehensive understanding of the environmental challenges. This information aids in devising effective mitigation strategies, enhancing disaster preparedness, and safeguarding both the natural environment and the local communities from the adverse effects of volcanic activity. Continued monitoring and assessment using satellite imagery remain crucial for proactive environmental management and sustainable development in the region.

As for the Stromboli region, it has been historically seen to have caused wildfires induced by the Volcano's eruption. There are also evident changes in temperatures that directly affect the vegetation and change in food supply, while increasing turbidity. The environment is altered after the volcano eruption. Aquatic life increases acidity. This isn't good for the fish and other animals because if the acidity enters their system, it harms them.

The climate changes by the mixture of particles and gases into the air. Some of these particles such as ash and sulfur dioxide (SO₂) have a cooling effect as they reflect sunlight. These particles, when released into the air, spread across the stratosphere and induce cooling effect during large eruptions as the sunlight is reflected back without reaching the earth's surface. Gases released like CO₂ cause warming by adding to the greenhouse effect.

There are not in the Stromboli region many victims as the residential population are about 850 people that make their home in two villages along the coast. However, there are recent updates on a tourist hiker being killed and Sentinel-2 Imagery of year 2019 and 2022 indicate the hot lava flow reaching the sea-level. Stromboli's most violent and destructive historic event occurred on 11, September 1930 that destroyed 14 houses in Ginostra, one of the two residential villages leaving behind 10 to 12 cm thick hot avalanches deposit and killing at least four people.

5 Conclusion

In summary, our study on mapping Stromboli's lava flows using Sentinel-2 imagery has provided valuable insights into the dynamic nature of volcanic activity on the island. Through meticulous analysis and advanced image processing techniques, we have achieved significant outcomes:

Accurate Spatial Mapping: Utilizing the high-resolution capabilities of Sentinel-2 imagery, we accurately mapped the spatial extent of Stromboli's lava flows. The detailed images allowed us to discern distinct flow patterns, providing a comprehensive overview of the affected areas.

Temporal Evolution Analysis: Our study enabled a temporal analysis of the lava flows. By comparing multiple satellite images taken over specific time intervals, we traced the evolution of lava flow fronts. This temporal perspective provided a deeper understanding of the dynamics and behavior of volcanic activity. The detailed lava flow maps generated in this study have crucial implications for risk assessment and mitigation strategies. By identifying vulnerable areas and predicting potential lava flow paths, our findings can aid local authorities and emergency responders in developing effective evacuation plans and minimizing the impact on nearby communities.

5.1 Future Research Directions

In future research, expanding the scope beyond Sentinel-2 data is essential to bridge the significant temporal gaps in our analysis. Integrating additional remote sensing datasets equipped with thermal bands holds promise for a more comprehensive understanding of environmental dynamics. By incorporating thermal infrared data from sources such as Landsat-8/9 Thermal Infrared Sensor (TIRS), Sentinel-3 Sea and Land Surface Temperature Radiometer (SLSTR), and Sentinel-1 Differential SAR interferometry (DInSAR) techniques we can overcome the limitations posed by large time intervals between observations.

This expanded approach will enable a holistic view of environmental changes, especially in areas where factors like land surface temperature, vegetation health, and urban heat islands play pivotal roles. A multi-sensor fusion strategy will be employed to effectively merge optical and thermal datasets. Advanced techniques, including machine learning algorithms and data fusion methods, will be explored to extract meaningful patterns and trends from the integrated datasets.

In the future, the research endeavors are directed towards the continuation of the study, employing advanced Big Data technology with a specific focus on leveraging Apache Hadoop's cluster environment. This approach facilitates parallel processing of a wide array of remote sensing and ground-based data, each characterized by distinct structural complexities. Notably, the research aims to harness the power of cloud-based data handling methods, adding a layer of flexibility and scalability to the analysis.

By integrating Big Data technology and employing parallel processing techniques, the study aspires to efficiently handle the diverse structures inherent in both remote sensing and ground-based data. This inclusivity not only aligns with the evolving trends in data science

but also addresses the multifaceted nature of data sources in the field. The utilization of Apache Hadoop's cluster environment allows for seamless parallel processing, enabling the research to delve into complex datasets simultaneously.

Furthermore, the incorporation of cloud data handling signifies a forward-looking approach, acknowledging the importance of scalable and adaptable solutions. By embracing cloud-based methodologies, the research endeavors can benefit from enhanced storage capacities, collaborative capabilities, and streamlined data processing workflows. This strategic integration is poised to unlock new dimensions in the analysis of remote sensing and ground data, leading to deeper insights and innovative discoveries in the field.

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7 Appendices

7.1 Sentinel-2 MSI Data Processing Code

Year	Month	Date	Hour	Minutes	Seconds	Time Interval	Whole Lava Flow area (m ²)	NHI tool Lava flow area (m ²)	Difference in Area in m ²	Accuracy Statement	Hot Area (m ²)
2019	1	8	10	26	52		9000	13200	-4200		4000
2019	1	8	11	26	13	1 hour	9600	14000	-4400	C	1200
										The output value of NHI tool has been clubbed to an 8000 m ² as an average for both the Sentinel-2 images with different ingestion times where as the Thesis	
2019	2	7	11	56	8		4000	8000	-4000		1600
2019	2	7	12	42	50	46.42 mins	7500	8000	-500		1500
2019	2	17	11	52	3		16700	18800	-2100	D	7600
										The output value of NHI tool has been clubbed to an 8000 m ² as an average for both the Sentinel-2 images with different ingestion times where as the Thesis	
2019	3	9	11	55	20		4800	6800	-2000		1200
2019	3	9	12	27	52	32.32 mins	7200	6800	400		2800
2019	3	24	13	34	30		5200	9200	-4000		1600
2019	3	24	14	20	46	46.16 mins	8000	9200	-1200	B	2400
2019	4	23	11	54	30		21200	24400	-3200		7600
2019	4	23	12	39	16	43.46 mins	34900			A	14800
2019	6	7	11	0	55		6400	6000	400		1600
2019	6	7	11	37	31	36.36 mins	8500	6000	2500	A	2000
2019	6	17	11	54	45		15600	14800	800	D	7600
2019	7	7	11	3	39		35600	36000	-400	D	16000
2019	7	7	11	42	0	38.39 mins	42300	41400	900	D	19200
2019	7	12	11	54	22		81000	98800	-17800		34400
2019	7	12	12	41	56	47.34 mins	125100	98800	27100	A	44000
2019	7	17	11	54	56		180400	200000	-19600		85600
2019	7	17	12	43	7	48.11 mins	277800	200000	77800	A	119200
2019	7	22	11	55	26		112100	150400	-38300		64100
2019	7	22	12	40	47	45.21 mins	132400	150400	-18000	A	78400
2019	8	1	11	55	45		80000	88800	-8800		41600
2019	8	1	12	30	52	35.07 mins	97600	88800	8800	A	48400
2019	8	6	11	3	3		109600	143200	-33600		48800
2019	8	6	11	42	40	39.37 mins	177200	143200	34000	A	73600
2019	8	11	12	20	17		75600	88800	-13200		39200
2019	8	11	12	55	24	35.07 mins	107000	88800	18800	A	51200
2019	8	21	11	54	26		9200	49600	-40400		1200
2019	8	21	12	33	0	38.34 mins	20600				3600
2019	9	5	11	3	59		12400	23600	-11200		6400
2019	9	5	11	47	58	43.59 mins	18200	23600	-5400	A	9800
2019	9	25	15	58	21		20400	21200	-800	D	5100
2019	10	15	10	24	39	1 hour, 3.43 mins	25200	38000	-12800		10000
2019	10	15	11	28	22		50900	39200	11700	A	15600
2019	10	20	11	58	36		26400	40800	-14400		4800
2019	10	20	12	37	40	39.4 mins	39600	41200	-1600	A	11200
2019	11	14	10	25	52	1 hour 2.34 mins	24400	48000	-23600		8000
2019	11	14	11	28	26		75400	48400	27000	A	19800
2019	12	14	10	24	14	1 hour 43.23 mins	26800	35600	-8800		14000
2019	12	14	12	7	37		50800	36000	14800	A	22400
Accuracy Statement Classification											
The tool has computed only single Occurance of Lava of the day where as from the Sentinel-2 it can be seen there is an increasing Lava flow between time intervals											
A -	The NHI tool uses Sentinel-2 and LandSat8 data for computation, which tends to variation in the resolution of The NHI algorithm detects the Lava area considering appropriate Thermal effects of Clouds, whereas the thesis work neglects the Cloud Interference.										
B -											
C -											
D -	Approximately similar outcome										

Table 3. 2019 Lava flow area Accuracy and Validation

Year	Month	Date	Hour	Minutes	Seconds	Time Interval	Whole	NHI tool	Differenc	Accuracy Statement	Sentinel-2	
							LavaFlow area (m ²)	Lava flow area (m ²)	in Area in m ²		Hot Area(m ²)	Cloud Percentage
2020	1	8	10	23	30	55.24 mins	14000	18800	-4800	E	8400	10.19
2020	1	8	11	18	54		20800	19600	1200		9900	14.49
2020	1	18	11	6	9	1 hour 19 secs	17500	34400	-16900	C	9900	31.09
2020	2	7	11	8	43		10000	11600	-1600		4000	1.5
2020	2	7	12	9	4		18200	17100	1100	D	7000	5.17
2020	2	17	11	50	35	42.52 mins	6200	12400	-6200		2700	1.1302
2020	2	17	12	33	23		15100	12400	2700	A	4000	2.006
2020	2	22	11	59	41	39.54 mins	16800	27600	-10800		6000	48.86
2020	2	22	12	39	35		37800	27600	10200	E	8300	52.638
2020	3	13	10	25	5	1 hour 2.21 mins	13800	18400	-4600		4800	17.42
2020	3	13	11	27	26		18000	18000	0	D	7000	24.04
2020	4	7	11	52	32	42.26 mins	10400	14000	-3600		2800	18.66
2020	4	7	12	34	58		13600	14000	-400	A	6400	26.526
2020	4	12	11	12	40	57.36 mins	19200	32800	-13600		6000	0.0309
2020	4	12	12	10	16		31600	32800	-1200	A, D	10400	0.236
2020	4	17	12	12	39	37.36 mins	8400	13200	-4800		2400	0
2020	4	17	12	50	15		12400	13200	-800	A, D	4400	0.4836
2020	5	7	11	55	11	48.30 mins	7200	8400	-1200		3200	29.43
2020	5	7	12	43	41		8400	8400	0	A, D	4000	37.06
2020	5	12	11	11	17	1 hour 20.22 mins	8400	8400	0		2000	0.7181
2020	5	12	12	31	39		9900	8400	1500	A, D	4000	1.961
2020	5	22	11	55	29	34.28 mins	5600	5600	0		1600	1.5889
2020	5	22	12	29	57		7200	5600	1600	A, B, D	2000	2.419
2020	5	27	11	55	7	36.10 mins	6000	8400	-2400		1600	2.8294
2020	5	27	12	31	17		9500	8400	1100	A, D	2000	10.925
2020	6	26	11	50	38	42.4 mins	4000	8000	-4000		1600	0.0222
2020	6	26	12	32	34		10400	8400	2000	A, E	2800	11.42
2020	7	1	11	58	56	37.1 mins	7600	9200	-1600		2000	0.306
2020	7	1	12	35	57		9600	9200	400	A, D	2800	11.35
2020	7	11	11	5	43	59.13 mins	7600	10400	-2800		2800	3.5166
2020	7	11	12	5	0		10000	10400	-400	A, D	2800	6.5357
2020	7	31	10	25	5	1 hour 1.41 mins	2500	2800	-300		400	0.2197
2020	7	31	11	26	46		2700	2800	-100	A, D	800	0.92
2020	8	15	11	5	10	1 hour 1.40 mins	6500	8000	-1500		1600	0.0096
2020	8	15	12	6	50		10000	8000	2000	A, E	3200	0.127
2020	8	20	12	3	57	36.55 mins	4000	8400	-4400		1200	0.3745
2020	8	20	12	40	52		7200	8400	-1200	A	1600	0.688
2020	8	30	11	7	31	46.54 mins	5600	11200	-5600		2400	1.1531
2020	8	30	11	54	23		11600	11200	400	A, D	4800	1.57
2020	9	9	11	4	15	1 hour 2.11 mins	3600	5600	-2000		2000	0.804
2020	9	9	12	6	26		5200	5600	-400	A, D	2400	1.3
2020	9	14	11	10	45	1 hour 2.58 mins	1600	6400	-4800		1200	0.0641
2020	9	14	12	13	43		3300	6400	-3100	A	500	4.68
2020	9	24	12	4	23	13 hours 1.47 mins	2000	6400	-4400		1600	2.59
2020	9	25	1	6	10		5500	6400	-900	A, D	2100	48.239
2020	10	4	11	2	34	1 hour 2.17 mins	4200	4000	200		1200	24.41
2020	10	4	12	4	17		3000	3600	-600	D	1800	30.18
2020	10	9	11	4	39	1 hour 6.6 mins	5200	10000	-4800		1600	9.47
2020	10	9	12	10	25		8400	9600	-1200	A, E	2800	3.11
2020	10	14	11	3	18		2800	3200	-400	D	600	21.75
2020	12	28	11	3	0		3700	4800	-1100	D	1200	2.91

Table 4. 2020 Lava flow area Accuracy and Validation

Year	Month	Date	Hour	Minutes	Seconds	Time Interval	Whole	NHI tool	Difference	Accuracy Statement	Sentinel-2	
							Lava Flow area (m ²)	Lava flow area (m ²)	in Area in m ²		Hot Area(m ²)	Cloud Percentage
2021	1	7	11	2	50	1 hour 31.36	6000	11600	-5600	A	3600	9.4209
2021	1	7	12	34	26	mins	9700	11600	-1900		4100	16.858
2021	2	6	11	4	16	58.39 mins	11600	14800	-3200	A	4800	0
2021	2	6	12	6	55		13300	14800	-1500		4700	0.0539
2021	2	21	10	55	6	56.43 mins	4700	8400	-3700	D	2700	2.78
2021	2	21	11	51	49		8700	8800	-100		4200	3.78
2021	3	3	10	51	37	42.58 mins	7900	12800	-4900	A, D	1900	0.5205
2021	3	3	11	34	35		12100	12800	-700		3600	1.227
2021	3	28	10	51	47		2800	3600	-800	D	800	0.1269
2021	5	7	11	1	19	3 hour 17.20	11600	12800	-1200	D	4000	31.938
2021	5	7	15	18	39	mins	12800	12800	0		4000	31.318
2021	5	17	11	1	45	1 hour 2.1	14800	18400	-3600	A, D	5200	8.8117
2021	5	17	12	3	44	mins	19200	18400	800		7200	11.439
2021	5	22	10	43	29	1 hour 50.59	139200	151200	-12000	A, E	66400	0.5367
2021	5	22	11	34	28	mins	220800	151600	69200		111200	1.679
2021	6	1	10	53	5	1 hour 25.37	13200	15300	-2100	B	7200	3.7681
2021	6	1	12	19	34	mins	17100	17200	-100	D	8800	7.07
2021	6	11	10	54	58	1 hour 25.45	15100	20000	-4900	A	8400	6.8228
2021	6	11	12	20	43	mins	23400	20000	3400		8800	9.254
2021	6	16	15	18	45	49.11 mins	16400	17600	-1200	A	7200	10.64
2021	6	16	16	9	56		19200	17600	1600		8400	16.9
2021	6	26	12	15	48		21600	18800	2800	E	8800	27.89
2021	7	1	11	4	25	1 hour 37.23	8800	6800	2000	A	1200	1.8446
2021	7	1	12	41	16	mins	10800	6800	4000		3200	3.733
2021	7	6	11	2	2	1 hour 18.51	2400	4400	-2000	A, D	1200	1.3845
2021	7	6	12	20	53	mins	4000	4400	-400		1600	7.494
2021	7	11	11	1	40	1 hour 17.19	5600	6800	-1200	A, D	1200	5.76
2021	7	11	12	18	59	mins	7200	6800	400		2800	8.202
2021	7	16	10	59	24	1 hour 3.34	4800	6000	-1200	A, D	2400	0.806
2021	7	16	12	2	58	mins	6400	6000	400		3200	2.63
2021	7	21	10	37	42	50.34 mins	8800	13600	-4800	A, F	2400	4.9506
2021	7	21	11	28	16		12500	13600	-1100		5200	6.864
2021	7	26	12	19	50	56.19 mins	6800	7600	-800	D	2000	0.2705
2021	7	26	13	16	9		10400	7200	3200	E	5600	1.198
2021	7	31	10	42	40	1 hour 22.46	4800	3600	1200	E	1600	0
2021	7	31	12	5	26	mins	6000	4000	2000		3600	0.066577
2021	8	10	10	46	22	1 hour 20.42	4400	5600	-1200	A	1200	0
2021	8	10	12	7	4	mins	8000	5600	2400		1600	0.0615
2021	8	25	11	0	17	1 hour 3.7	8000	9600	-1600	A, D	3200	0.753
2021	8	25	12	3	28	mins	10400	9600	800		4400	2.433
2021	8	30	11	35	52		7600	9200	-1600	D	3600	6.136
2021	9	19	12	56	34		6400	6800	-400	D	2000	0.0517
2021	9	24	11	24	19		3200	5600	-2400	B	2000	4.97
2021	10	4	10	51	11	1 hour 20.13	4400	8800	-4400	A, D	1600	0.2939
2021	10	4	12	11	24	mins	8800	8800	0		1600	0.46
2021	11	3	10	50	49	1 hour 26.5	13600	13600	0	D	6400	4.15
2021	11	3	12	16	54	mins	17200	14000	3200	E	10900	5.367
2021	12	8	12	18	14		15800	25200	-9400	C	5400	11.176
2021	12	18	11	37	24		13200	10400	2800	B, E	9800	57.48
2021	12	23	10	52	34	1 hour 29.36	3600	6800	-3200	C	1600	5.19

Table 5. 2021 Lava flow area Accuracy and Validation

Year	Month	Date	Hour	Minutes	Seconds	Time Interval	Whole	NHI tool	Difference	Accuracy Statement	Hot	Sentinel-2
							Lava Flow area (m ²)	Lava flow area (m ²)	in Area in m ²		Area(m ²)	Cloud Percentage
2022	1	12	12	8	17		5300	7200	-1900	B, F	2400	7.62
2022	1	22	12	0	52	59.9 mins	6800	8000	-1200	A	4000	36.55
2022	1	22	12	57	1		9500	8000	1500		5400	36.78
2022	2	11	12	9	35	1 hour 15.15 mins	6400	10800	-4400	A, B	2400	0
2022	2	11	13	24	50		11200	10800	400		4400	0.028
2022	4	12	10	54	19	1 hour 22.6 mins	10500	11600	-1100	A, D	4800	27.39
2022	4	12	12	16	25		12300	11600	700		5500	17.57
2022	4	22	10	56	12	2 hours 21.47 mins	5200	6000	-800	A, D	2400	2.23
2022	4	22	13	6	38		7600	6000	1600		2400	3.96
2022	5	2	10	40	38	3 hours 25.42 mins	2000	2400	-400	A, D	400	4.8358
2022	5	2	14	6	20		2800	2400	400		1200	6.91
2022	5	12	11	50	2	2 hours 49.13 mins	1600	1800	-200	A, B, D	0	2.009
2022	5	12	14	39	15		2400	3200	-800		0	3.47
2022	5	22	11	54	3	2 hours 31.1 mins	1200	1200	0	A, D	0	0
2022	5	22	14	25	4		2400	1200	1200		800	1.192
2022	6	1	11	52	27	2 hours 33.44 mins	6800	7600	-800	A, D	3200	0
2022	6	1	14	26	11		7200	7600	-400		4400	4.96
2022	6	11	11	52	8		3600	4000	-400	D, E	1200	7.969
2022	6	21	11	50	9		2000	2400	-400	D, E	800	0.0105
2022	7	1	11	50	2		1600	1600	0	D	400	11.323
2022	7	11	11	51	13		800	400	400	D	0	2.686
2022	7	31	11	50	36	2 hours 28.21 mins	6000	6800	-800	B, E	2800	11.006
2022	7	31	14	18	57		9900	6800	3100		4000	17.325
2022	8	25	14	18	55		7600	6400	1200	D	3200	51.39
2022	9	9	13	30	27	2 hours 26.30 mins	3600			B	1600	2.433
2022	9	9	15	56	57		5000	6400	-1400		2400	3.68
2022	9	14	12	38	30		9000	10400	-1400	D	2700	0.0209
2022	9	19	11	55	14		6400	10800	-4400	C	2800	7.486
2022	10	4	12	9	14		23600	39200	-15600	E	12800	8.599
2022	10	9	11	50	20		140800	182400	-41600	C	33600	48.829
2022	10	24	10	45	22		4000	4800	-800	D	1600	0.938
2022	11	8	11	53	27	2 hours 32.43 mins	9200	10800	-1600	C	4000	2.305
2022	11	8	14	26	0		9700			A	2700	5.017
2022	11	13	14	24	52		14500	18000	-3500	C	4300	37.887
2022	12	8	11	50	12	2 hours 22.45 mins	96100	140800	-44700	A, C, D	52500	4.523
2022	12	8	14	12	57		132900	144800	-11900		103700	9.015
2022	12	28	14	11	0		60400	39200	21200	E	15900	7.86
Accuracy Statement Classification												
A - The tool has computed only single Occurance of Lava of the computation, which tends to variation in the resolution of pixels												
B - The NHI algorithm detects the Lava area considering appropriate Thermal effects of Clouds and Ash fumes, whereas the thesis work neglects the Cloud Interference												
C - Approximately similar outcome												
D - tool, Thesis work seems to precisely pick the lava area in this particular scenario, and only the resolution considered by NHI algorithm explains the difference in Area												
E - The images does not seem to differ between the computed work and NHI tool, however there is a difference in area												
F -												

Table 6. 2022 Lava flow area Accuracy and Validation

Year	Month	Date	Hour	Minutes	Seconds	Time Interval	Whole Lava Flow area (m ²)	NHI tool Lava flow area (m ²)	Difference in Area in m ²	Accuracy Statement	Hot Area(m ²)
2023	2	1	11	31	37		14100	11600	2500	E	6000
2023	2	11	10	42	18	45.57 mins	8400	10800	-2400	E	3600
2023	2	11	11	28	15		12000	10400	1600	E	6000
2023	3	13	20	24	55	1 hour 24.5 mins	8800	9600	-800	A, D	3200
2023	3	13	21	49	0		10800	9600	1200		5200
2023	3	23	15	9	21		12600	20000	-7400	E	6800
2023	3	28	13	57	1		11800	12400	-600	B, D	6300
2023	4	27	14	4	12		5600	6800	-1200	A	800
2023	5	7	11	51	49	2 hours 26.3 mins	4000	5600	-1600	F, D	1600
2023	5	7	14	17	52		6400	6000	400		1600
2023	6	1	14	31	34		800	3200	-2400	C	0
2023	6	21	10	38	56	51.52 mins	6400	8000	-1600	E	2400
2023	6	21	11	30	44		9200	7600	1600		3200
Accuracy Statement Classification											
The tool has computed only single Occurance of Lava of the day whereas from the Sentinel-2 it can be seen there is an increasing Lava flow between time intervals computation, which tends to variation in the resolution of pixels											
A - appropriate Thermal effects of Clouds and Ash fumes, whereas the thesis work neglects the Cloud Interference.											
B - Approximately similar outcome											
C - tool, Thesis work seems to precisely pick the lava area in this particular scenario, and only the resolution considered by The images does not seem to differ between the computed work and NHI tool, however there is a difference in area											
D -											
E -											
F -											

Table 7. 2022 Lava flow area Accuracy and Validation

7.2 Acknowledgements

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